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1. Setup and Utilities

Template

```
1 #include "bits/stdc++.h"
2 using namespace std;
3 using ll = long long;
4 #define int ll
5 #define endl '\n'
6 void solve() {
7 }
8 signed main() {
9     cin.tie(0)->sync_with_stdio(0);
10    cin.exceptions(cin.failbit);
11    int tt = 1;
12    // cin >> tt;
13    while (tt--) {
14        solve();
15        cout << '\n';
16    }
17    return 0;
18 }
```

Precompile stdc++.h

```
1 sudo g++ -x c++-header -std=c++17 -O0 stdc++.h -o stdc++.h.gch
2 ulimit -s ${size in kb}
3 add empty template to live templates
4 increase undo range to 10k (ctrl+shift+A) -> registry -> undo
```

Pragmas

```
1 #pragma GCC optimize("Ofast")
2 #pragma GCC optimize ("unroll-loops")
3 #pragma GCC target("sse,sse2,sse3,ssse3,sse4,popcnt,abm,mmx,avx,tune=native")
```

Random mt19937

```
1 mt19937 rng(chrono::steady_clock::now().time_since_epoch().count());
2 ll rnd(ll l, ll r) {
3     static mt19937_64 gen(chrono::steady_clock::now().time_since_epoch().count());
4     return uniform_int_distribution<ll>(l, r)(gen);
5 }
```

Direction Arrays

```
1 int dx[8] = { 2, 1, -1, -2, -2, -1, 1, 2 };
2 int dy[8] = { 1, 2, 2, 1, -1, -2, -2, -1 }; // knight
3
4 int dx[8] = {-1,0,1,-1,1,-1,0,1};
5 int dy[8] = {-1,-1,-1,0,0,1,1,1}; // king
6
7 int dx[4] = {1, -1, 0, 0};
8 int dy[4] = {0, 0, -1, 1};
9 string direction = "DULR";
```

Custom Comparator

```
1 struct cmp {
2     bool operator() (int a, int b) const {
3         return ...;
4     }
5 };
6 set<int, cmp> s;
```

Fast HashMap (gp_hash_table)

```
1 #include <ext/pb_ds/assoc_container.hpp>
2 using namespace __gnu_pbds;
3 struct chash {
4     const int RANDOM = (ll)(make_unique<char>().get()) ^
5         chrono::high_resolution_clock::now().time_since_epoch().count();
6
7     static unsigned long long hash_f(unsigned long long x) {
8         x += 0x9e3779b97f4a7c15;
9         x = (x ^ (x >> 30)) * 0xbf58476d1ce4e5b9;
10        x = (x ^ (x >> 27)) * 0x94d049bb133111eb;
11        return x ^ (x >> 31);
12    }
13    static unsigned hash_combine(unsigned a, unsigned b) {
14        return a * 31 + b;
15    }
16    int operator()(int x) const {
17        return hash_f(x)^RANDOM;
18    }
19 };
20 gp_hash_table<int, int, chash> table;
```

__int128_t

```
1 istream& operator>>(istream& is, __int128_t& v) {
2     string s; is >> s; v = 0;
3     for (char c : s) if (isdigit(c)) v = v * 10 + c - '0';
4     if (s[0] == '-') v = -v;
5     return is;
6 }
7 ostream& operator<<(ostream& os, __int128_t v) {
8     if (!v) return os << "0";
9     if (v < 0) os << '-', v = -v;
10    string s;
11    while (v) s += '0' + v % 10, v /= 10;
12    reverse(s.begin(), s.end());
13    return os << s;
14 }
```

2. Data Structures

Mo struct

```
1  const int N = 1e5 + 5, sq = 317;
2  struct query{
3      int l, r, i;
4      bool operator<(const query &other) const {
5          if (l / sq != other.l / sq)
6              return l / sq < other.l / sq;
7          return (l / sq & 1 ? r < other.r : other.r < r);
8      }
9  };
```

Mo with updates

```
1  const int N = 2e5 + 5;
2  int a[N];
3  vector<int> coords;
4  long long curAns, ans[N];
5
6  struct Query {
7      int l, r, t, id, blk_l, blk_r;
8
9      bool operator<(const Query& o) const {
10         if (blk_l != o.blk_l) return blk_l < o.blk_l;
11         if (blk_r != o.blk_r) return (blk_l & 1) ? blk_r < o.blk_r : blk_r > o.blk_r;
12         return (blk_r & 1) ? t < o.t : t > o.t;
13     }
14 };
15
16 struct Update {
17     int p, v;
18 };
19
20 vector<Query> queries;
21 vector<Update> updates;
22 int vals[N];
23
24 inline void add(int i) {
25     if (++vals[a[i]] == 1) curAns += coords[a[i]];
26 }
27
28 inline void del(int i) {
29     if (!--vals[a[i]]) curAns -= coords[a[i]];
30 }
31
32 inline void do_update(int t, int l, int r) {
33     auto &u = updates[t];
34     if (l <= u.p && u.p <= r) del(u.p);
35     swap(a[u.p], u.v);
36     if (l <= u.p && u.p <= r) add(u.p);
37 }
38
39 void process(int n) {
40     constexpr int block = 700;
41     for (auto &q: queries) {
```

```

42         q.blk_l = q.l / block;
43         q.blk_r = q.r / block;
44     }
45     sort(queries.begin(), queries.end());
46
47     int l = 1, r = 0, t = 0;
48     for (const auto &q: queries) {
49         while (t < q.t) do_update(t++, l, r);
50         while (t > q.t) do_update(--t, l, r);
51         while (l > q.l) add(--l);
52         while (r < q.r) add(++r);
53         while (l < q.l) del(l++);
54         while (r > q.r) del(r--);
55
56         ans[q.id] = curAns;
57     }
58 }
59
60 void solve() {
61     int n;
62     cin >> n;
63     coords.reserve(n << 1);
64     for (int i = 0; i < n; i++) cin >> a[i], coords.push_back(a[i]);
65     int q, j = 0;
66     cin >> q;
67     for (int i = 0; i < q; i++) {
68         string s;
69         cin >> s;
70         if (s.front() == 'Q') {
71             int l, r;
72             cin >> l >> r;
73             queries.push_back({--l, --r, (int) updates.size(), j, 0, 0});
74             j++;
75         } else {
76             int idx, val;
77             cin >> idx >> val;
78             --idx;
79             updates.push_back({idx, val});
80             coords.push_back(val);
81         }
82     }
83
84     sort(coords.begin(), coords.end());
85     coords.erase(unique(coords.begin(), coords.end()), coords.end());
86     for (int i = 0; i < n; i++) a[i] = lower_bound(coords.begin(), coords.end(), a[i]) - coords.begin();
87     for (auto &o: updates) o.v = lower_bound(coords.begin(), coords.end(), o.v) - coords.begin();
88
89     // pass n to the process function
90     process(n);
91
92     for (int i = 0; i < j; i++) cout << ans[i] << ' ';
93
94     queries.clear();
95     updates.clear();
96     coords.clear();
97     curAns = 0;
98 }

```

2D Prefix Sum

```
1 // construction
2 for (int i = 1; i <= N; i++) {
3     for (int j = 1; j <= N; j++) {
4         prefix[i][j] =
5             arr[i][j]
6             + prefix[i - 1][j]
7             + prefix[i][j - 1]
8             - prefix[i - 1][j - 1];
9     }
10 }
11
12 // query
13 pfx[to_row][to_col]
14 - pfx[from_row - 1][to_col]
15 - pfx[to_row][from_col - 1]
16 + pfx[from_row - 1][from_col - 1];
17
18 // partial: add value v to subrectangle [from_row, from_col] to [to_row, to_col]
19 diff[from_row][from_col] += v;
20 diff[from_row][to_col + 1] -= v;
21 diff[to_row + 1][from_col] -= v;
22 diff[to_row + 1][to_col + 1] += v;
23 // then construct diff, grid[i][j] += diff[i][j];
```

Multiset Lazy deletion

```
1 template<typename T, typename Compare = less<T>>
2 struct MS {
3     priority_queue<T, vector<T>, Compare> pq, del;
4     void normalize(){
5         while(!pq.empty() && !del.empty() && pq.top() == del.top()){
6             pq.pop();
7             del.pop();
8         }
9     }
10    bool empty() { normalize(); return size() == 0; }
11    int size() { return (int)pq.size() - (int)del.size(); }
12    void insert(T x) { pq.push(x); }
13    void erase(T x) { del.push(x); }
14    T top() { normalize(); return pq.top(); }
15    void pop() { normalize(); pq.pop(); }
16    void clear() {
17        while(!pq.empty()) pq.pop();
18        while(!del.empty()) del.pop();
19    }
20 };
```

BIT, Fenwick Tree

```
1 struct BIT { // 0-based
2     int n;
3     vector<ll> tree;
4
5     BIT(int size) : n(size + 2), tree(n + 1) {}
```



```

6 void add(int i, ll val) {
7     for(i++; i <= n; i += i & -i) tree[i] += val;
8 }
9 ll query(int i) {
10     ll sum = 0;
11     for (i++; i > 0; i &= i - 1) sum += tree[i];
12     return sum;
13 }
14 int lower_bound(ll target) {
15     int i = 0;
16     ll curr = 0;
17     for (int mask = 1 << __lg(n); mask > 0; mask >>= 1) {
18         if (i + mask <= n && curr + tree[i + mask] < target) {
19             curr += tree[i += mask];
20         }
21     }
22     return i;
23 }
24 };

```

BIT Range / Fenwick Range

```

1 template<typename T>
2 class BitR { // 0-based
3     int n;
4     vector<T> f, s;
5     void add(vector<T> &a, int i, T val) {
6         for(; i < n; i += i & -i)
7             a[i] += val;
8     }
9 public:
10     BitR(int n) : n(n + 5), f(n + 6), s(n + 6) { }
11
12     void add(int i, T val) { add(s, i + 1, -val); }
13
14     T query_point(int i) {
15         if (!i) return query(0);
16         return query(i) - query(i - 1);
17     }
18     void set(int i, T val) {
19         int c = query_point(i);
20         add(i, val - c);
21     }
22
23     void add(int l, int r, T val) {
24         l++, r++;
25         add(f, l, val);
26         add(f, r + 1, -val);
27         add(s, l, val * (l - 1));
28         add(s, r + 1, -val * r);
29     }
30
31     T query(int ii) {
32         ii++;
33         T sum = 0;
34         int i = ii;

```

```

35         for(; i > 0; i ^= i & -i)
36             sum += f[i];
37         sum *= ii;
38         i = ii;
39         for(; i > 0; i ^= i & -i)
40             sum -= s[i];
41         return sum;
42     }
43
44     T range(int l, int r) { return query(r) - query(l - 1); }
45 };

```

2D BIT / 2D fenwick

```

1  template<typename T>
2  struct BIT2D { // 1-based
3      int n, m;
4      vector<vector<T>>> tree;
5
6      BIT2D(int n, int m) : n(n), m(m), tree(n + 2, vector<T>(m + 2, 0)) {}
7
8      void update(int x, int y, T val) {
9          for (int i = x; i <= n; i += i & -i) {
10             for (int j = y; j <= m; j += j & -j) {
11                 tree[i][j] += val;
12             }
13         }
14     }
15
16     T getPrefix(int x, int y) {
17         if (x <= 0 || y <= 0) return 0;
18         T ret = 0; // change default value
19         for (int i = x; i > 0; i &= i - 1) {
20             for (int j = y; j > 0; j &= j - 1) {
21                 ret += tree[i][j];
22             }
23         }
24         return ret;
25     }
26
27     T getSquare(int xl, int yl, int xr, int yr) { // change operation
28         return getPrefix(xr, yr) + getPrefix(xl - 1, yl - 1) -
29             getPrefix(xr, yl - 1) - getPrefix(xl - 1, yr);
30     }
31 };

```

Segment Tree

```

1  struct info {
2      long long sum = 0;
3      info(long long sum = 0) : sum(sum) {}
4      friend info operator+(const info &l, const info &r) {
5          return {l.sum + r.sum};
6      }
7  };
8

```

```

9  template<class info>
10 struct segmentTreeIterative {
11     int n;
12     vector<info> tree;
13
14     explicit segmentTreeIterative(int n) : n(n), tree(n << 1) {}
15
16     template<class U>
17     explicit segmentTreeIterative(const vector<U> &arr) : n(arr.size()),
18         tree(n << 1) {
19         for(int i = 0; i < n; i++) tree[i + n] = info(arr[i]);
20         for(int i = n - 1; i > 0; i--) tree[i] = tree[i << 1] + tree[i << 1 | 1];
21     }
22
23     void set(int i, info v) {
24         for(tree[i += n] = v; i >>= 1; )
25             tree[i] = tree[i << 1] + tree[i << 1 | 1];
26     }
27
28     info get(int l, int r) {
29         info resL, resR;
30         for(l += n, r += n + 1; l < r; l >>= 1, r >>= 1) {
31             if(l & 1) resL = resL + tree[l++];
32             if(r & 1) resR = tree[--r] + resR;
33         }
34         return resL + resR;
35     }
36 };

```

Segment Tree (Recursive)

```

1  struct info {
2      long long sum = 0;
3      info(long long sum = 0) : sum(sum) {}
4      friend info operator+(const info &l, const info &r) {
5          return {l.sum + r.sum};
6      }
7  };
8
9  template<class info>
10 struct dynamicSegmentTree {
11     struct node { int l = 0, r = 0; info v; };
12     vector<node> tr;
13     int n, root = 0;
14
15     explicit dynamicSegmentTree(int n = 1e9, int expectedOps = 1) : n(n), tr(1) {
16         tr.reserve(expectedOps * __lg(n) * 2);
17     }
18
19     info get(int x, int lx, int rx, int l, int r) {
20         if (!x || lx > r || l > rx) return info();
21         if (lx >= l && rx <= r) return tr[x].v;
22         int m = (lx + rx) >> 1;
23         return get(tr[x].l, lx, m, l, r) + get(tr[x].r, m + 1, rx, l, r);
24     }
25

```

```

26     int set(int x, int lx, int rx, int i, info val) {
27         if (i < lx || i > rx) return x;
28         if (!x) x = tr.size(), tr.emplace_back();
29         if (lx == rx) return tr[x].v = val, x;
30         int m = (lx + rx) >> 1;
31         tr[x].l = set(tr[x].l, lx, m, i, val);
32         tr[x].r = set(tr[x].r, m + 1, rx, i, val);
33         return tr[x].v = tr[tr[x].l].v + tr[tr[x].r].v, x;
34     }
35
36     void set(int i, info v) { root = set(root, 0, n, i, v); }
37     info get(int l, int r) { return get(root, 0, n, l, r); }
38 };

```

2D Segment Tree

```

1  struct info {
2      long long sum = 0;
3      info(long long sum = 0) : sum(sum) {}
4      friend info operator+(const info &l, const info &r) {
5          return {l.sum + r.sum};
6      }
7  };
8
9  template<class info>
10 struct segmentTree2d {
11     int nx, ny;
12     vector<vector<info>> tree;
13
14     explicit segmentTree2d(int n, int m) : nx(n), ny(m), tree(nx << 1,
15         vector<info>(ny << 1)) {}
16
17     template<class U>
18     explicit segmentTree2d(const vector<vector<U>> &a) {
19         nx = a.size();
20         ny = nx ? a[0].size() : 0;
21         tree.assign(nx << 1, vector<info>(ny << 1));
22
23         for(int i = 0; i < nx; i++) {
24             for(int j = 0; j < ny; j++) tree[i + nx][j + ny] = info(a[i][j]);
25             for(int y = ny - 1; y > 0; y--)
26                 tree[i + nx][y] = tree[i + nx][y << 1] + tree[i + nx][y << 1 | 1];
27         }
28         for(int x = nx - 1; x > 0; x--)
29             for(int y = 0; y < (ny << 1); y++)
30                 tree[x][y] = tree[x << 1][y] + tree[x << 1 | 1][y];
31     }
32
33     void set(int i, int j, info v) {
34         if (i < 0 || i >= nx || j < 0 || j >= ny) return;
35         int x = i + nx, y = j + ny;
36         tree[x][y] = v;
37         for(int yy = y >> 1; yy > 0; yy >>= 1)
38             tree[x][yy] = tree[x][yy << 1] + tree[x][yy << 1 | 1];
39
40         for(int xx = x >> 1; xx > 0; xx >>= 1) {

```

```

41         tree[xx][y] = tree[xx << 1][y] + tree[xx << 1 | 1][y];
42         for(int yy = y >> 1; yy > 0; yy >>= 1)
43             tree[xx][yy] = tree[xx][yy << 1] + tree[xx][yy << 1 | 1];
44     }
45 }
46
47 info queryY(int nodeX, int y1, int y2) const {
48     info resL, resR;
49     for(int l = y1 + ny, r = y2 + ny + 1; l < r; l >>= 1, r >>= 1) {
50         if (l & 1) resL = resL + tree[nodeX][l++];
51         if (r & 1) resR = tree[nodeX][--r] + resR;
52     }
53     return resL + resR;
54 }
55
56 info get(int x1, int y1, int x2, int y2) {
57     if (!nx || !ny) return info();
58     x1 = max(x1, 0); y1 = max(y1, 0);
59     x2 = min(x2, nx - 1); y2 = min(y2, ny - 1);
60     if (x1 > x2 || y1 > y2) return info();
61
62     info resL, resR;
63     for(int l = x1 + nx, r = x2 + nx + 1; l < r; l >>= 1, r >>= 1) {
64         if (l & 1) resL = resL + queryY(l++, y1, y2);
65         if (r & 1) resR = queryY(--r, y1, y2) + resR;
66     }
67     return resL + resR;
68 }
69 };

```

Lazy Segment Tree

```

1  struct tag {
2      ll add = 0;
3      void apply(const tag &p) {
4          add += p.add;
5      }
6  };
7
8  struct info {
9      ll sum = 0;
10     info(ll sum = 0) : sum(sum) {}
11     void apply(const tag &lz, int lx, int rx) {
12         sum += 1ll * (rx - lx + 1) * lz.add;
13     }
14     friend info operator+(const info &l, const info &r) {
15         return {l.sum + r.sum};
16     }
17 };
18
19 template<class info, class tag>
20 struct lazySegment {
21     int n;
22     vector<info> tr;
23     vector<tag> lz;
24
25     explicit lazySegment(int _n) : n(_n + 1), tr(4 << __lg(n)), lz(4 << __lg(n)) {}

```

```

26     template<class U>
27     explicit lazySegment(const U &arr) : n(arr.size()), tr(4 << __lg(n)),
28         lz(4 << __lg(n)) {
29         build(1, 0, n - 1, arr);
30     }
31
32     void push(int x, int lx, int rx) {
33         if (lx != rx) lz[x << 1].apply(lz[x]), lz[x << 1 | 1].apply(lz[x]);
34         tr[x].apply(lz[x], lx, rx);
35         lz[x] = tag();
36     }
37
38     template<class U>
39     void build(int x, int lx, int rx, const U &arr) {
40         if (lx == rx) return void(tr[x] = arr[lx]);
41         int m = (lx + rx) >> 1;
42         build(x << 1, lx, m, arr), build(x << 1 | 1, m + 1, rx, arr);
43         tr[x] = tr[x << 1] + tr[x << 1 | 1];
44     }
45
46     info get(int x, int lx, int rx, int l, int r) {
47         push(x, lx, rx);
48         if (lx > r || l > rx) return info();
49         if (lx >= l && rx <= r) return tr[x];
50         int m = (lx + rx) >> 1;
51         return get(x << 1, lx, m, l, r) + get(x << 1 | 1, m + 1, rx, l, r);
52     }
53
54     void set(int x, int lx, int rx, int i, info val) {
55         push(x, lx, rx);
56         if (i < lx || i > rx) return;
57         if (lx == rx) return void(tr[x] = val);
58         int m = (lx + rx) >> 1;
59         set(x << 1, lx, m, i, val), set(x << 1 | 1, m + 1, rx, i, val);
60         tr[x] = tr[x << 1] + tr[x << 1 | 1];
61     }
62
63     void applyRange(int x, int lx, int rx, int l, int r, tag v) {
64         push(x, lx, rx);
65         if (lx > r || l > rx) return;
66         if (lx >= l && rx <= r) return lz[x] = v, push(x, lx, rx);
67         int m = (lx + rx) >> 1;
68         applyRange(x << 1, lx, m, l, r, v),
69         applyRange(x << 1 | 1, m + 1, rx, l, r, v);
70         tr[x] = tr[x << 1] + tr[x << 1 | 1];
71     }
72
73     void set(int i, info v) { set(1, 0, n - 1, i, v); }
74     void applyRange(int l, int r, tag v) { applyRange(1, 0, n - 1, l, r, v); }
75     info get(int l, int r) { return get(1, 0, n - 1, l, r); }
76 };

```

Dynamic Lazy Segment Tree

```

1  template<class info, class tag>

```

```

2 struct dynamicLazySegmentTree {
3     struct node { int l = 0, r = 0; info v; tag t; };
4     vector<node> tr;
5     int n, root = 0;
6
7     explicit dynamicLazySegmentTree(int n = 1e9) : n(n), tr(1) {}
8
9     void push(int x, int lx, int rx) {
10         if (lx == rx) return void(tr[x].t = tag());
11         if (!tr[x].l) tr[x].l = tr.size(), tr.emplace_back();
12         if (!tr[x].r) tr[x].r = tr.size(), tr.emplace_back();
13
14         int m = (lx + rx) >> 1, l = tr[x].l, r = tr[x].r;
15         tr[l].t.apply(tr[x].t), tr[l].v.apply(tr[x].t, lx, m);
16         tr[r].t.apply(tr[x].t), tr[r].v.apply(tr[x].t, m + 1, rx);
17         tr[x].t = tag();
18     }
19
20     int set(int x, int lx, int rx, int i, info val) {
21         if (i < lx || i > rx) return x;
22         if (!x) x = tr.size(), tr.emplace_back();
23         if (lx == rx) {
24             tr[x].v = val, tr[x].t = tag();
25             return x;
26         }
27         push(x, lx, rx);
28         int m = (lx + rx) >> 1;
29         tr[x].l = set(tr[x].l, lx, m, i, val);
30         tr[x].r = set(tr[x].r, m + 1, rx, i, val);
31         return tr[x].v = tr[tr[x].l].v + tr[tr[x].r].v, x;
32     }
33
34     int applyRange(int x, int lx, int rx, int l, int r, tag val) {
35         if (lx > r || l > rx) return x;
36         if (!x) x = tr.size(), tr.emplace_back();
37         if (lx >= l && rx <= r) {
38             tr[x].t.apply(val);
39             tr[x].v.apply(val, lx, rx);
40             return x;
41         }
42         push(x, lx, rx);
43         int m = (lx + rx) >> 1;
44         tr[x].l = applyRange(tr[x].l, lx, m, l, r, val);
45         tr[x].r = applyRange(tr[x].r, m + 1, rx, l, r, val);
46         return tr[x].v = tr[tr[x].l].v + tr[tr[x].r].v, x;
47     }
48
49     info get(int x, int lx, int rx, int l, int r) {
50         if (!x || lx > r || l > rx) return info();
51         if (lx >= l && rx <= r) return tr[x].v;
52         push(x, lx, rx);
53         int m = (lx + rx) >> 1;
54         return get(tr[x].l, lx, m, l, r) + get(tr[x].r, m + 1, rx, l, r);
55     }
56
57     void set(int i, info v) { root = set(root, 0, n, i, v); }

```

```

58 void applyRange(int l, int r, tag v) { root = applyRange(root, 0, n, l, r, v); }
59 info get(int l, int r) { return get(root, 0, n, l, r); }
60 };

```

Segment Tree Beats

```

1 struct segtreebeats {
2     static const ll INF = 2e18, UNSET = LLONG_MIN;
3
4     struct Node {
5         ll sum, mx1, mx2, mxc, mn1, mn2, mnc, d_gcd, lz_add, lz_set;
6     };
7
8     int sz;
9     vector<Node> tree;
10
11     Node leaf(ll v) { return {v, v, -INF, 1, v, INF, 1, 0, 0, UNSET}; }
12
13     segtreebeats(int n, const vector<ll> &a) {
14         for (sz = 1; sz < n; sz <= 1);
15         tree.assign(sz < 1, leaf(0));
16         build(a, n, 0, 0, sz - 1);
17     }
18
19     Node merge(const Node &L, const Node &R) {
20         Node U = {L.sum + R.sum, 0, 0, 0, 0, 0, 0, gcd(L.d_gcd, R.d_gcd), 0, UNSET};
21
22         if (L.mx1 == R.mx1) U.mx1 = L.mx1, U.mx2 = max(L.mx2, R.mx2), U.mxc = L.mxc + R.mxc;
23         else if (L.mx1 > R.mx1) U.mx1 = L.mx1, U.mx2 = max(L.mx2, R.mx1), U.mxc = L.mxc;
24         else U.mx1 = R.mx1, U.mx2 = max(L.mx1, R.mx2), U.mxc = R.mxc;
25
26         if (L.mn1 == R.mn1) U.mn1 = L.mn1, U.mn2 = min(L.mn2, R.mn2), U.mnc = L.mnc + R.mnc;
27         else if (L.mn1 < R.mn1) U.mn1 = L.mn1, U.mn2 = min(L.mn2, R.mn1), U.mnc = L.mnc;
28         else U.mn1 = R.mn1, U.mn2 = min(L.mn1, R.mn2), U.mnc = R.mnc;
29
30         ll aL = L.mx2, aR = R.mx2;
31         if (aL != -INF && aL != L.mn1 && aR != -INF && aR != R.mn1)
32             U.d_gcd = gcd(U.d_gcd, abs(aL - aR));
33
34         ll any = UNSET;
35         if (aL != -INF && aL != L.mn1) any = aL;
36         else if (aR != -INF && aR != R.mn1) any = aR;
37
38         for (ll val : {L.mn1, L.mx1, R.mn1, R.mx1}) {
39             if (val != U.mn1 && val != U.mx1) {
40                 if (any != UNSET) U.d_gcd = gcd(U.d_gcd, abs(val - any));
41                 else any = val;
42             }
43         }
44         return U;
45     }
46
47     void apply_set(int x, int lx, int rx, ll v) {
48         ll len = rx - lx + 1;
49         tree[x] = {len * v, v, -INF, len, v, INF, len, 0, 0, v};
50     }

```



```

51
52 void apply_add(int x, int lx, int rx, ll v) {
53     if (!v) return;
54     auto &nd = tree[x];
55     if (nd.lz_set != UNSET) return apply_set(x, lx, rx, nd.lz_set + v);
56     if (nd.mx1 == nd.mn1) return apply_set(x, lx, rx, nd.mn1 + v);
57     ll len = rx - lx + 1;
58     nd.sum += len * v;
59     nd.mx1 += v; if (nd.mx2 != -INF) nd.mx2 += v;
60     nd.mn1 += v; if (nd.mn2 != INF) nd.mn2 += v;
61     nd.lz_add += v;
62 }
63
64 void apply_chmin(int x, int lx, int rx, ll v) {
65     auto &nd = tree[x];
66     if (nd.mx1 <= v) return;
67     if (nd.mn1 >= v) return apply_set(x, lx, rx, v);
68     if (nd.mn2 == nd.mx1) nd.mn2 = v;
69     nd.sum -= (nd.mx1 - v) * nd.mxc;
70     nd.mx1 = v;
71 }
72
73 void apply_chmax(int x, int lx, int rx, ll v) {
74     auto &nd = tree[x];
75     if (nd.mn1 >= v) return;
76     if (nd.mx1 <= v) return apply_set(x, lx, rx, v);
77     if (nd.mx2 == nd.mn1) nd.mx2 = v;
78     nd.sum += (v - nd.mn1) * nd.mnc;
79     nd.mn1 = v;
80 }
81
82 void push(int x, int lx, int rx) {
83     if (lx == rx) return tree[x].lz_add = 0, tree[x].lz_set = UNSET, void();
84     int m = (lx + rx) >> 1, lc = x * 2 + 1, rc = x * 2 + 2;
85
86     if (tree[x].lz_set != UNSET) {
87         apply_set(lc, lx, m, tree[x].lz_set);
88         apply_set(rc, m + 1, rx, tree[x].lz_set);
89         tree[x].lz_set = UNSET;
90     }
91     if (tree[x].lz_add) {
92         apply_add(lc, lx, m, tree[x].lz_add);
93         apply_add(rc, m + 1, rx, tree[x].lz_add);
94         tree[x].lz_add = 0;
95     }
96     if (tree[lc].mx1 > tree[x].mx1) apply_chmin(lc, lx, m, tree[x].mx1);
97     if (tree[rc].mx1 > tree[x].mx1) apply_chmin(rc, m + 1, rx, tree[x].mx1);
98     if (tree[lc].mn1 < tree[x].mn1) apply_chmax(lc, lx, m, tree[x].mn1);
99     if (tree[rc].mn1 < tree[x].mn1) apply_chmax(rc, m + 1, rx, tree[x].mn1);
100 }
101
102 void build(const vector<ll> &a, int n, int x, int lx, int rx) {
103     if (lx == rx) return void(tree[x] = (lx < n) ? leaf(a[lx]) : leaf(0));
104     int m = (lx + rx) >> 1;
105     build(a, n, x * 2 + 1, lx, m);
106     build(a, n, x * 2 + 2, m + 1, rx);

```

```

107     tree[x] = merge(tree[x * 2 + 1], tree[x * 2 + 2]);
108 }
109
110 void chmin(int l, int r, ll v, int x, int lx, int rx) {
111     if (lx > r || rx < l || tree[x].mx1 <= v) return;
112     if (lx >= l && rx <= r && tree[x].mx2 < v) return apply_chmin(x, lx, rx, v);
113     push(x, lx, rx);
114     int m = (lx + rx) >> 1;
115     chmin(l, r, v, x * 2 + 1, lx, m);
116     chmin(l, r, v, x * 2 + 2, m + 1, rx);
117     tree[x] = merge(tree[x * 2 + 1], tree[x * 2 + 2]);
118 }
119
120 void chmax(int l, int r, ll v, int x, int lx, int rx) {
121     if (lx > r || rx < l || tree[x].mn1 >= v) return;
122     if (lx >= l && rx <= r && tree[x].mn2 > v) return apply_chmax(x, lx, rx, v);
123     push(x, lx, rx);
124     int m = (lx + rx) >> 1;
125     chmax(l, r, v, x * 2 + 1, lx, m);
126     chmax(l, r, v, x * 2 + 2, m + 1, rx);
127     tree[x] = merge(tree[x * 2 + 1], tree[x * 2 + 2]);
128 }
129
130 void assign(int l, int r, ll v, int x, int lx, int rx) {
131     if (lx > r || rx < l) return;
132     if (lx >= l && rx <= r) return apply_set(x, lx, rx, v);
133     push(x, lx, rx);
134     int m = (lx + rx) >> 1;
135     assign(l, r, v, x * 2 + 1, lx, m);
136     assign(l, r, v, x * 2 + 2, m + 1, rx);
137     tree[x] = merge(tree[x * 2 + 1], tree[x * 2 + 2]);
138 }
139
140 void add(int l, int r, ll v, int x, int lx, int rx) {
141     if (lx > r || rx < l) return;
142     if (lx >= l && rx <= r) return apply_add(x, lx, rx, v);
143     push(x, lx, rx);
144     int m = (lx + rx) >> 1;
145     add(l, r, v, x * 2 + 1, lx, m);
146     add(l, r, v, x * 2 + 2, m + 1, rx);
147     tree[x] = merge(tree[x * 2 + 1], tree[x * 2 + 2]);
148 }
149
150 ll sum(int l, int r, int x, int lx, int rx) {
151     if (lx > r || rx < l) return 0;
152     if (lx >= l && rx <= r) return tree[x].sum;
153     push(x, lx, rx);
154     int m = (lx + rx) >> 1;
155     return sum(l, r, x * 2 + 1, lx, m) + sum(l, r, x * 2 + 2, m + 1, rx);
156 }
157
158 ll qmin(int l, int r, int x, int lx, int rx) {
159     if (lx > r || rx < l) return INF;
160     if (lx >= l && rx <= r) return tree[x].mn1;
161     push(x, lx, rx);

```

```

162         int m = (lx + rx) >> 1;
163         return min(qmin(l, r, x * 2 + 1, lx, m), qmin(l, r, x * 2 + 2, m + 1, rx));
164     }
165
166     ll qmax(int l, int r, int x, int lx, int rx) {
167         if (lx > r || rx < l) return -INF;
168         if (lx >= l && rx <= r) return tree[x].mx1;
169         push(x, lx, rx);
170         int m = (lx + rx) >> 1;
171         return max(qmax(l, r, x * 2 + 1, lx, m), qmax(l, r, x * 2 + 2, m + 1, rx));
172     }
173
174     ll qgcd(int l, int r, int x, int lx, int rx) {
175         if (lx > r || rx < l) return 0;
176         if (lx >= l && rx <= r) {
177             ll ans = gcd(tree[x].d_gcd, abs(tree[x].mx1));
178             if (tree[x].mx2 != -INF) ans = gcd(ans, abs(tree[x].mx2 - tree[x].mx1));
179             if (tree[x].mn2 != INF) ans = gcd(ans, abs(tree[x].mn2 - tree[x].mn1));
180             return ans;
181         }
182         push(x, lx, rx);
183         int m = (lx + rx) >> 1;
184         return gcd(qgcd(l, r, x * 2 + 1, lx, m), qgcd(l, r, x * 2 + 2, m + 1, rx));
185     }
186
187     // Public Wrappers
188     void chmin(int l, int r, ll v) { chmin(l, r, v, 0, 0, sz - 1); }
189     void chmax(int l, int r, ll v) { chmax(l, r, v, 0, 0, sz - 1); }
190     void assign(int l, int r, ll v) { assign(l, r, v, 0, 0, sz - 1); }
191     void add(int l, int r, ll v) { add(l, r, v, 0, 0, sz - 1); }
192     ll qsum(int l, int r) { return sum(l, r, 0, 0, sz - 1); }
193     ll qmin(int l, int r) { return qmin(l, r, 0, 0, sz - 1); }
194     ll qmax(int l, int r) { return qmax(l, r, 0, 0, sz - 1); }
195     ll qgcd(int l, int r) { return qgcd(l, r, 0, 0, sz - 1); }
196 };

```

Dynamic Persistent Segment Tree

```

1  template<class info>
2  struct DPSGT {
3      struct node { int l = 0, r = 0; info v; };
4      vector<node> tr;
5      int n;
6
7      explicit DPSGT(int n = 1e9, int expectedOps = 1) : n(n), tr(1) {
8          tr.reserve(expectedOps * __lg(n) * 2);
9      }
10
11     info get(int x, int lx, int rx, int l, int r) {
12         if (!x || lx > r || l > rx) return info();
13         if (lx >= l && rx <= r) return tr[x].v;
14         int m = (lx + rx) >> 1;
15         return get(tr[x].l, lx, m, l, r) + get(tr[x].r, m + 1, rx, l, r);
16     }
17
18     int set(int x, int lx, int rx, int i, info val) {

```

```

19         if (i < lx || i > rx) return x;
20         int nx = tr.size();
21         tr.push_back(tr[x]);
22         if (lx == rx) return tr[nx].v = val, nx;
23         int m = (lx + rx) >> 1;
24         tr[nx].l = set(tr[nx].l, lx, m, i, val);
25         tr[nx].r = set(tr[nx].r, m + 1, rx, i, val);
26         return tr[nx].v = tr[tr[nx].l].v + tr[tr[nx].r].v, nx;
27     }
28
29     int set(int root, int i, info v) { return set(root, 0, n, i, v); }
30     info get(int root, int l, int r) { return get(root, 0, n, l, r); }
31 };

```

Wavelet Tree

```

1  struct WaveletTree {
2      int lo, hi;
3      WaveletTree *lc = 0, *rc = 0;
4      vector<int> b;
5
6      template<class It>
7      WaveletTree(It from, It to, int x, int y) : lo(x), hi(y) {
8          if (from >= to || lo == hi) return;
9          int mid = lo + (hi - lo) / 2;
10
11          b.reserve(distance(from, to) + 1);
12          b.push_back(0);
13          for (auto it = from; it != to; ++it) {
14              b.push_back(b.back() + (*it <= mid));
15          }
16
17          auto pivot = stable_partition(from, to, [mid](int val)
18              { return val <= mid; });
19          lc = new WaveletTree(from, pivot, lo, mid);
20          rc = new WaveletTree(pivot, to, mid + 1, hi);
21      }
22
23      // the passed vector is changed, pass a copy in the main
24      // Takes a vector of integers, and the min/max values in it.
25      // The vector should be 0-indexed but queries works as 1-based
26      WaveletTree(vector<int>& a, int x, int y) :
27      WaveletTree(a.begin(), a.end(), x, y) {}
28
29      ~WaveletTree() { delete lc; delete rc; }
30
31      int kth(int l, int r, int k) {
32          if (l > r) return 0;
33          if (lo == hi) return lo;
34          int in_left = b[r] - b[l - 1], lb = b[l - 1], rb = b[r];
35          if (k <= in_left) return lc->kth(lb + 1, rb, k);
36          return rc->kth(l - lb, r - rb, k - in_left);
37      }
38
39      int LTE(int l, int r, int k) {
40          if (l > r || k < lo) return 0;

```

```

41         if (hi <= k) return r - l + 1;
42         int lb = b[l - 1], rb = b[r];
43         return lc->LTE(lb + 1, rb, k) + rc->LTE(l - lb, r - rb, k);
44     }
45
46     int count(int l, int r, int k) {
47         if (l > r || k < lo || k > hi) return 0;
48         if (lo == hi) return r - l + 1;
49         int mid = lo + (hi - lo) / 2, lb = b[l - 1], rb = b[r];
50         if (k <= mid) return lc->count(lb + 1, rb, k);
51         return rc->count(l - lb, r - rb, k);
52     }
53 };

```

Implicit Treap

```

1 // You must manually track the root of your tree in your main function:
2 //     Treap tree;
3 //     int root = 0;
4 // 1. Insert val at index i:
5 //     int l, r;
6 //     tree.split(root, i, l, r);
7 //     int mid = tree.new_node(val);
8 //     root = tree.merge(tree.merge(l, mid), r);
9 //
10 // 2. Delete the element at index i:
11 //     auto [l, mid, r] = tree.split(root, i, i);
12 //     root = tree.merge(l, r); // mid is safely dropped and ignored
13 //
14 // 3. Range Sum for subarray [L, R]:
15 //     auto [l, mid, r] = tree.split(root, L, R);
16 //     long long ans = tree.tr[mid].sum;
17 //     root = tree.merge(tree.merge(l, mid), r); // ALWAYS merge back!
18 //
19 // 4. Range Reverse for subarray [L, R]:
20 //     auto [l, mid, r] = tree.split(root, L, R);
21 //     tree.tr[mid].rev ^= 1; // apply the lazy tag
22 //     root = tree.merge(tree.merge(l, mid), r); // ALWAYS merge back!
23 // -----
24
25 mt19937 rnd(time(nullptr));
26 struct Treap {
27     struct node {
28         uint32_t pri = rnd();
29         int sz = 1, l = 0, r = 0, val{};
30         int64_t sum{};
31         bool rev = false;
32         node(int x) : sum(val = x) { }
33     };
34     vector<node> tr;
35
36     // tr[0] acts as a dummy/null node to avoid segmentation faults
37     Treap() : tr(1, 0) { tr[0].sz = 0; }
38
39     inline void pull(int x) {
40         tr[x].sz = tr[tr[x].l].sz + tr[tr[x].r].sz + 1;

```

```

41     tr[x].sum = tr[tr[x].l].sum + tr[tr[x].r].sum + tr[x].val;
42 }
43
44 inline void push(int x) {
45     if(tr[x].rev) {
46         tr[tr[x].l].rev ^= 1, tr[tr[x].r].rev ^= 1;
47         swap(tr[x].l, tr[x].r);
48         tr[x].rev = false;
49     }
50 }
51
52 // Returns the index of the newly created node
53 inline int new_node(int val) {
54     tr.emplace_back(val);
55     return int(tr.size()) - 1;
56 }
57
58 // Concatenates treap rx to the right of lx. Returns the new root.
59 int merge(int lx, int rx) {
60     if(!lx || !rx) return rx + lx;
61     push(lx), push(rx);
62     if(tr[lx].pri < tr[rx].pri)
63         return tr[rx].l = merge(lx, tr[rx].l), pull(rx), rx;
64     return tr[lx].r = merge(tr[lx].r, rx), pull(lx), lx;
65 }
66
67 // Splits treap x into lx (first k elements) and rx (the rest).
68 void split(int x, int k, int &lx, int &rx) {
69     if(!x) return lx = rx = 0, void();
70     push(x);
71     if(k <= tr[tr[x].l].sz) split(tr[x].l, k, lx, tr[x].l), pull(rx = x);
72     else split(tr[x].r, k - tr[tr[x].l].sz - 1, tr[x].r, rx), pull(lx = x);
73 }
74
75 // Helper to extract a specific subarray [l, r] into the middle partition
76 array<int, 3> split(int x, int l, int r) {
77     int a, b, c;
78     split(x, r + 1, b, c);
79     split(b, l, a, b);
80     return {a, b, c};
81 }
82 };

```

Sparse Table

```

1  struct ST {
2      static const int K = 18, N = 2e5 + 5;
3      int st[K][N];
4
5      void build(int n, const vector<int>& a) {
6          for(int i = 0; i < n; i++) st[0][i] = a[i];
7          for(int k = 1; k < K; k++)
8              for(int i = 0; i + (1 << k) <= n; i++)
9                  st[k][i] = min(st[k - 1][i], st[k - 1][i + (1 << (k - 1))]);
10     }
11 }

```

```

12     int query(int l, int r) {
13         int k = __lg(r - l + 1);
14         return min(st[k][l], st[k][r - (1 << k) + 1]);
15     }
16 };
17
18
19 template<class T, class F>
20 struct sparse {
21     int n, Log;
22     vector<vector<T>> table;
23     F merge;
24     T id;
25
26     sparse(const vector<T>& arr, F merge, T id = T()) :
27         n(arr.size()), Log(__lg(n) + 1), table(Log, vector<T>(n)),
28         merge(merge), id(id) {
29         table[0] = arr;
30         for (int l = 1; l < Log; l++) {
31             for (int i = 0; i + (1 << (l - 1)) < n; i++) {
32                 table[l][i] = merge(table[l - 1][i],
33                                     table[l - 1][i + (1 << (l - 1))]);
34             }
35         }
36     }
37
38     T query(int l, int r) {
39         if (l > r) return id;
40         int len = __lg(r - l + 1);
41         return merge(table[len][l], table[len][r - (1 << len) + 1]);
42     }
43
44     T query_log(int l, int r) {
45         T res = id;
46         bool first = true;
47         for (int j = Log - 1; j >= 0; j--) {
48             if (1 << j <= r - l + 1) {
49                 res = first ? table[j][l] : merge(res, table[j][l]);
50                 first = false;
51                 l += 1 << j;
52             }
53         }
54         return res;
55     }
56 };
57 // sparse st(a, [](int x, int y) { return min(x, y); }, INT_MAX);

```

2D Sparse Table

```

1  #include <bits/stdc++.h>
2  using namespace std;
3
4  template<class T, class F>
5  struct sparse2d {
6      int n, m, Kx, Ky;
7      vector<int> lgx, lgy;
8      vector<vector<vector<vector<T>>>> st;

```

```

9      F merge;
10
11      sparse2d(const vector<vector<T>>& a, F merge) : merge(merge) {
12          n = (int)a.size();
13          m = (int)a[0].size();
14
15          lgx.resize(n + 1);
16          lgy.resize(m + 1);
17          for (int i = 2; i <= n; ++i) lgx[i] = lgx[i / 2] + 1;
18          for (int j = 2; j <= m; ++j) lgy[j] = lgy[j / 2] + 1;
19
20          Kx = lgx[n];
21          Ky = lgy[m];
22
23          st.assign(Kx + 1, vector<vector<vector<T>>>(Ky + 1));
24
25          st[0][0] = a;
26
27          // Build along columns for kx = 0
28          for (int ky = 1; ky <= Ky; ++ky) {
29              int len = 1 << ky;
30              int half = len >> 1;
31              st[0][ky].assign(n, vector<T>(m - len + 1));
32              for (int i = 0; i < n; ++i) {
33                  for (int j = 0; j + len <= m; ++j) {
34                      st[0][ky][i][j] = merge(st[0][ky - 1][i][j],
35                                              st[0][ky - 1][i][j + half]);
36                  }
37              }
38          }
39
40          // Build along rows for all ky
41          for (int kx = 1; kx <= Kx; ++kx) {
42              int lenx = 1 << kx;
43              int halfx = lenx >> 1;
44              for (int ky = 0; ky <= Ky; ++ky) {
45                  int leny = 1 << ky;
46                  st[kx][ky].assign(n - lenx + 1, vector<T>(m - leny + 1));
47                  for (int i = 0; i + lenx <= n; ++i) {
48                      for (int j = 0; j + leny <= m; ++j) {
49                          st[kx][ky][i][j] = merge(st[kx - 1][ky][i][j],
50                                                  st[kx - 1][ky][i + halfx][j]);
51                      }
52                  }
53              }
54          }
55      }
56
57      // inclusive coordinates: [x1..x2], [y1..y2]
58      T query(int x1, int y1, int x2, int y2) const {
59          int kx = lgx[x2 - x1 + 1];
60          int ky = lgy[y2 - y1 + 1];
61          int dx = 1 << kx;
62          int dy = 1 << ky;
63
64          const T& a = st[kx][ky][x1][y1];
65          const T& b = st[kx][ky][x2 - dx + 1][y1];

```



```

66         const T& c = st[kx][ky][x1][y2 - dy + 1];
67         const T& d = st[kx][ky][x2 - dx + 1][y2 - dy + 1];
68
69         return merge(merge(a, b), merge(c, d));
70     }
71 };

```

DSU

```

1  struct DSU {
2      vector<int> p, sz;
3      int n, comps;
4      DSU(int _n = 0) { init(_n); }
5      void init(int _n) {
6          n = _n + 10; comps = _n;
7          p.resize(n); sz.assign(n, 1);
8          iota(p.begin(), p.end(), 0);
9      }
10     int find(int u) { return u == p[u] ? u : p[u] = find(p[u]); }
11     bool unite(int u, int v) {
12         u = find(u), v = find(v);
13         if (u == v) return 0;
14         if (sz[u] < sz[v]) swap(u, v);
15         p[v] = u; sz[u] += sz[v]; comps--;
16         return 1;
17     }
18     bool same(int u, int v) { return find(u) == find(v); }
19     int size(int u) { return sz[find(u)]; }
20     int size() { return comps; }
21 };

```

Persistent DSU

```

1  struct persistent_DSU {
2      vector<vector<pair<int, int> > > par;
3      persistent_DSU(int n) : par(n + 1, {{-1, 0}}) {}
4
5      int time = 0;
6      bool merge(int u, int v) {
7          ++time;
8          if ((u = root(u, time)) == (v = root(v, time)))
9              return 0;
10         if (par[u].back().first > par[v].back().first)
11             swap(u, v);
12         par[u].push_back({par[u].back().first + par[v].back().first, time});
13         par[v].push_back({u, time}); // par[v] = u
14         return 1;
15     }
16
17     bool same(int u, int v, int t) {
18         return root(u, t) == root(v, t);
19     }
20
21     int root(int u, int t) {
22         // root of u at time t

```

```

23         if (par[u].back().first >= 0 && par[u].back().second <= t)
24             return root(par[u].back().first, t);
25         return u;
26     }
27
28     int size(int u, int t) { // O(log)
29         // size of the component of u at time t
30         u = root(u, t);
31         int l = 0, r = (int) par[u].size() - 1, ans = 0;
32         while (l <= r) {
33             int mid = l + r >> 1;
34             if (par[u][mid].second <= t)
35                 ans = mid, l = mid + 1;
36             else
37                 r = mid - 1;
38         }
39         return -par[u][ans].first;
40     }
41 };

```

Bipartite DSU

```

1 // Maintains whether each component is bipartite
2 struct BipartiteDSU {
3     vector<int> sz, bipartite;
4     vector<pair<int, int> > par;
5
6     BipartiteDSU(int n) : par(n), sz(n, 1), bipartite(n) {
7         for (int i = 0; i < n; ++i) {
8             par[i] = {i, 0};
9         }
10    }
11
12    pair<int, int> find(int u) {
13        if (u == par[u].first) return {u, 0};
14        int parity = par[u].second;
15        par[u] = find(par[u].first);
16        par[u].second ^= parity;
17        return par[u];
18    }
19
20    bool same(int x, int y) {
21        return find(x).first == find(y).first;
22    }
23
24    bool join(int u, int v) {
25        pair<int, int> pu = find(u);
26        pair<int, int> pv = find(v);
27        u = pu.first;
28        v = pv.first;
29        int x = pu.second, y = pv.second;
30        if (u == v) {
31            if (x == y) bipartite[u] = false;
32            return false;
33        }
34        if (sz[u] < sz[v]) swap(u, v);
35        par[v] = {u, x ^ y ^ 1};

```

```

36         bipartite[u] &= bipartite[v];
37         sz[u] += sz[v];
38         return true;
39     }
40
41     int size(int x) { return sz[find(x).first]; }
42 };

```

Monotonic Stack / Queue

```

1  template<class T>
2  struct Mono_stack {
3      stack<pair<T, T> > st;
4
5      void push(const T &val) {
6          if (st.empty()) st.emplace(val, val);
7          else st.emplace(val, max(val, st.top().second));
8      }
9
10     void pop() { st.pop(); }
11     bool empty() { return st.empty(); }
12     int size() { return st.size(); }
13     T top() { return st.top().first; }
14     T max() { return st.top().second; }
15 };
16
17 template<class T>
18 struct Mono_queue {
19     Mono_stack<T> pop_st, push_st;
20
21     void push(const T &val) { push_st.push(val); }
22
23     void move() {
24         if (pop_st.size()) return;
25         while (!push_st.empty()) pop_st.push(push_st.top()), push_st.pop();
26     }
27
28     void pop() { move(); pop_st.pop(); }
29     bool empty() { return pop_st.empty() && push_st.empty(); }
30     int size() { return pop_st.size() + push_st.size(); }
31     T top() { move(); return pop_st.top(); }
32
33     T max() {
34         if (pop_st.empty()) return push_st.max();
35         if (push_st.empty()) return pop_st.max();
36         return max(push_st.max(), pop_st.max());
37     }
38 };

```

Ordered Data Structures (pb_ds)

```

1  #include <ext/pb_ds/assoc_container.hpp>
2  #include <ext/pb_ds/tree_policy.hpp>
3  using namespace __gnu_pbds;
4  template <typename T> using ordered_set = tree<T, null_type, less<T>,
5      rb_tree_tag, tree_order_statistics_node_update>;

```

```

6 template <typename T, typename R> using ordered_map = tree<T, R, less<T>,
7     rb_tree_tag, tree_order_statistics_node_update>;

```

BucketList

```

1  template<class T>
2  struct BucketList {
3      int siz = 0;
4      vector<vector<T> > a;
5      static constexpr int SPLIT_RATIO = 28;
6
7      void insert(int i, T x) {
8          if (siz == 0) {
9              siz = 1;
10             a = {{x}};
11             return;
12         }
13         siz++;
14         for (ll bi = 0; bi < size(a); bi++) {
15             auto &bucket = a[bi];
16             if (i <= size(bucket)) {
17                 bucket.insert(begin(bucket) + i, x);
18                 if (size(bucket) > size(a) * SPLIT_RATIO) {
19                     auto L = end(bucket) - size(bucket) / 2, R = end(bucket);
20                     a.emplace(begin(a) + bi + 1, L, R);
21                     // bucket might be broken
22                     a[bi].erase(L, R);
23                 }
24                 return;
25             }
26             i -= size(bucket);
27         }
28     }
29
30     void erase(int i) {
31         siz--;
32         for (int bi = 0; bi < size(a); bi++) {
33             auto& bucket = a[bi];
34             if (i < size(bucket)) {
35                 bucket.erase(begin(bucket) + i);
36                 if (bucket.empty()) a.erase(begin(a) + bi);
37                 return;
38             }
39             i -= size(bucket);
40         }
41     }
42
43     T& access(int i) {
44         for (auto& bucket : a) {
45             if (i < size(bucket)) return bucket[i];
46             i -= size(bucket);
47         }
48         assert(false);
49     }
50 };

```

Full Dynamic Array

```
1  mt19937 rng(chrono::steady_clock::now().time_since_epoch().count());
2  template<typename T>
3  struct DynArray {
4      struct Node {
5          T val; int pri, sz;
6          Node *l, *r;
7          Node(const T& v) : val(v), pri(rng()), sz(1), l(0), r(0) {}
8      };
9
10     Node* root = 0;
11
12     static int sz(Node* t) { return t ? t->sz : 0; }
13     static void pull(Node* t) { if (t) t->sz = 1 + sz(t->l) + sz(t->r); }
14
15     static void split(Node* t, int k, Node*& l, Node*& r) {
16         if (!t) { l = r = 0; return; }
17         if (sz(t->l) < k) {
18             split(t->r, k - sz(t->l) - 1, t->r, r);
19             l = t;
20         } else {
21             split(t->l, k, l, t->l);
22             r = t;
23         }
24         pull(t);
25     }
26
27     static void merge(Node*& t, Node* l, Node* r) {
28         if (!l || !r) { t = l ? l : r; return; }
29         if (l->pri > r->pri) {
30             merge(l->r, l->r, r);
31             t = l;
32         } else {
33             merge(r->l, l, r->l);
34             t = r;
35         }
36         pull(t);
37     }
38
39     void insert(int pos, const T& val) {
40         Node *l, *r;
41         split(root, pos, l, r);
42         Node* nd = new Node(val);
43         merge(l, l, nd);
44         merge(root, l, r);
45     }
46
47     void erase(int pos) {
48         Node *l, *m, *r;
49         split(root, pos, l, m);
50         split(m, 1, m, r);
51         delete m;
52         merge(root, l, r);
53     }
54
55     T& access(int pos) {
```

```

56     Node* t = root;
57     while (true) {
58         int left = sz(t->l);
59         if (pos < left) t = t->l;
60         else if (pos == left) return t->val;
61         else { pos -= left + 1; t = t->r; }
62     }
63 }
64
65 int size() { return sz(root); }
66 };

```

Count below

```

1 // counts how many pairs that sum <= limit
2 template <typename T>
3 ll count_below(vector<T>& v, int sz, ll Limit){
4     // unique pairs such that v[i]+v[j] <= limit
5     assert(is_sorted(v.begin(), v.end()));
6     ll total = 0;
7     for (int l = 0, r = sz-1; l < r; l++){
8         while(r > l && v[l] + v[r] > Limit)
9             r--;
10        total += max(0, r-l);
11    }
12    return total;
13 }

```

3. Graphs and Trees

Bellman Ford

```
1 // Computes Single-Source Shortest Paths (SSSP) on graphs that contain NEGATIVE edge weights.
2 // It explicitly detects negative weight cycles reachable from the source vertex.
3 // - Average Time Complexity:  $O(E)$  where  $E$  is the number of edges.
4 // - Worst Case Time Complexity:  $O(V * E)$ .
5 // vector<int> dist;
6 // bool ok = spfa(0, adj, dist);
7
8 const int INF = 1e9;
9 bool spfa(int s, const vector<vector<pair<int, int>>>& adj, vector<int>& d) {
10     int n = adj.size();
11     d.assign(n, INF);
12     vector<int> cnt(n, 0);
13     vector<int> inqueue(n, 0);
14     queue<int> q;
15
16     d[s] = 0;
17     q.push(s);
18     inqueue[s] = 1;
19
20     while (!q.empty()) {
21         int v = q.front();
22         q.pop();
23         inqueue[v] = 0;
24         for (const auto& [to, len] : adj[v]) {
25             if (d[v] + len < d[to]) {
26                 d[to] = d[v] + len;
27                 if (!inqueue[to]) {
28                     q.push(to);
29                     inqueue[to] = 1;
30                     if (++cnt[to] > n) return false; // negative cycle
31                 }
32             }
33         }
34     }
35     return true;
36 }
```

Floyd Tricks

```
1 void TransitiveClosure(int n, vector<vector<int>>& adj) {
2     // 0 = disconnected, 1 = connected
3     for (int k = 0; k < n; ++k)
4         for (int i = 0; i < n; ++i)
5             for (int j = 0; j < n; ++j)
6                 adj[i][j] |= (adj[i][k] & adj[k][j]);
7 }
8
9 void minimax(int n, vector<vector<int>>& adj) {
10     // Path such that max value on road is minimized
11     for (int k = 0; k < n; ++k)
12         for (int i = 0; i < n; ++i)
13             for (int j = 0; j < n; ++j)
```

```

14         adj[i][j] = min(adj[i][j], max(adj[i][k], adj[k][j]));
15     }
16
17 void maximin(int n, vector<vector<int>>& adj) {
18     // Path such that min value on road is maximized
19     for (int k = 0; k < n; ++k)
20         for (int i = 0; i < n; ++i)
21             for (int j = 0; j < n; ++j)
22                 adj[i][j] = max(adj[i][j], min(adj[i][k], adj[k][j]));
23 }
24
25 void longestPathDAG(int n, vector<vector<int>>& adj) {
26     // Only works for DAGs (no positive cycles)
27     for (int k = 0; k < n; ++k)
28         for (int i = 0; i < n; ++i)
29             for (int j = 0; j < n; ++j)
30                 adj[i][j] = max(adj[i][j], max(adj[i][k], adj[k][j]));
31 }
32
33 void countPaths(int n, vector<vector<int>>& adj) {
34     // Floyd-Warshall for counting number of paths
35     for (int k = 0; k < n; ++k)
36         for (int i = 0; i < n; ++i)
37             for (int j = 0; j < n; ++j)
38                 adj[i][j] += adj[i][k] * adj[k][j];
39 }
40
41 bool isNegativeCycle(int n, const vector<vector<int>>& adj) {
42     // run floyd first
43     for (int i = 0; i < n; ++i)
44         if (adj[i][i] < 0)
45             return true;
46     return false;
47 }
48
49 bool anyEffectiveCycle(int n, const vector<vector<int>>& adj,
50     int src, int dest, int 00) {
51     // run floyd first
52     for (int i = 0; i < n; ++i)
53         if (adj[i][i] < 0 && adj[src][i] < 00 && adj[i][dest] < 00)
54             return true;
55     return false;
56 }

```

Topological Sort

```

1  queue<int> queue;
2  for (int i = 0; i < n; i++)
3      if (in_degree[i] == 0) { queue.push(i); }
4
5  vector<int> top_sort;
6  while (!queue.empty()) {
7      int curr = queue.front();
8      queue.pop();
9      top_sort.push_back(curr);
10     for (int next : graph[curr]) {

```



```

11         if (--in_degree[next] == 0) { queue.push(next); }
12     }
13 }

```

Bipartite Check

```

1  int n;
2  vector<vector<int>> adj;
3
4  vector<int> side(n, -1);
5  bool is_bipartite = true;
6  queue<int> q;
7  for (int st = 0; st < n; ++st) {
8      if (side[st] == -1) {
9          q.push(st);
10         side[st] = 0;
11         while (!q.empty()) {
12             int v = q.front();
13             q.pop();
14             for (int u : adj[v]) {
15                 if (side[u] == -1) {
16                     side[u] = side[v] ^ 1;
17                     q.push(u);
18                 } else {
19                     is_bipartite &= side[u] != side[v];
20                 }
21             }
22         }
23     }
24 }

```

Dijkstra

```

1  vector<ll> dijkstra(int src, vector<int> &parent) {
2      vector<ll> dist(n + 1, INF);
3      parent.assign(n + 1, -1);
4      priority_queue<pair<ll, int>, vector<pair<ll, int> >, greater<pair<ll, int> > >
5          pq;
6      dist[src] = 0;
7      pq.push({0, src});
8      while (!pq.empty()) {
9          auto [d, v] = pq.top();
10         pq.pop();
11         if (d != dist[v]) continue;
12         for (auto [u, w]: graph[v]) {
13             if (dist[u] > d + w) {
14                 dist[u] = d + w;
15                 parent[u] = v;
16                 pq.push({dist[u], u});
17             }
18         }
19     }
20     return dist;
21 }

```

Tree Diameter

```
1 pair<int, int> dfs(int u, int p) {
2     pair<int, int> ret = {0, u};
3     for (int v : tree[u]) {
4         if (v == p) continue;
5         auto x = dfs(v, u);
6         ret = max(ret, {x.first + 1, x.second});
7     }
8     return ret;
9 }
```

Tree

```
1 struct tree {
2     int root = 0;
3     vector<vector<int>> g;
4     explicit tree(int n) : g(n) { }
5     void add(int u, int v) {
6         g[u].push_back(v);
7         g[v].push_back(u);
8     }
9
10    vector<int> &operator[](int u) {
11        return g[u];
12    }
13
14    int cntDfs = 0;
15    vector<int> in, out, lvl, sz, top, par, seq;
16
17    void init(int rt = 0) {
18        root = rt;
19        in = out = lvl = top = par = seq = vector<int>(g.size());
20        sz.resize(g.size(), 1);
21        par[root] = top[root] = root;
22        dfs(root);
23        dfs2(root);
24    }
25
26    void dfs(int u) {
27        for(int &v : g[u]) {
28            lvl[v] = lvl[u] + 1;
29            par[v] = u;
30            g[v].erase(find(g[v].begin(), g[v].end(), u));
31            dfs(v);
32            sz[u] += sz[v];
33            if(sz[v] > sz[g[u][0]])
34                swap(v, g[u][0]);
35        }
36    }
37    void dfs2(int u) {
38        in[u] = cntDfs++;
39        seq[in[u]] = u;
40        for(int v : g[u]) {
41            top[v] = v == g[u][0]? top[u]: v;
42            dfs2(v);
43        }
44    }
```

```

44     out[u] = cntDfs - 1;
45 }
46
47 int jump(int u, int k) {
48     if(k > lvl[u]) return -1;
49
50     int d = lvl[u] - k;
51     while (lvl[top[u]] > d) {
52         u = par[top[u]];
53     }
54
55     return seq[in[u] - lvl[u] + d];
56 }
57
58 bool isAncestor(int u, int v) {
59     return in[u] <= in[v] && in[v] <= out[u];
60 }
61
62 int lca(int u, int v) {
63     if(lvl[u] > lvl[v]) swap(u, v);
64     if(isAncestor(u, v)) return u;
65     while (top[u] != top[v]) {
66         if (lvl[top[u]] > lvl[top[v]]) {
67             u = par[top[u]];
68         } else {
69             v = par[top[v]];
70         }
71     }
72     return lvl[u] < lvl[v]? u: v;
73 }
74
75 int dis(int u, int v) {
76     return lvl[u] + lvl[v] - 2 * lvl[lca(u, v)];
77 }
78 };

```

Tree Hash (rooted/unrooted)

```

1  // tree hash
2  using u64 = uint64_t;
3  mt19937_64 rng = []{
4      u64 time_entropy = chrono::steady_clock::now().time_since_epoch().count();
5      u64 memory_entropy = (uintptr_t)make_unique<char>().get();
6      seed_seq ss{time_entropy, memory_entropy};
7      return mt19937_64(ss);
8  }();
9  u64 SEED = rng();
10 u64 mix(u64 x) {
11     x += SEED + 0x9e3779b97f4a7c15;
12     x = (x ^ x >> 30) * 0xbf58476d1ce4e5b9;
13     x = (x ^ x >> 27) * 0x94d049bb133111eb;
14     return x ^ x >> 31;
15 }
16 u64 treehash(int u, int p, vector<vector<int>>& t) {
17     u64 ret = 1;
18     // do ret = ret * P + mix() if the order is important

```

```

19     for (int v : t[u]) if (v ^ p)
20         ret += mix(treehash(v, u, t));
21     return ret;
22 }
23 // unrooted: find centroids and treat them as roots
24 u64 utreehash(vector<vector<int>>& t) {
25     int n = t.size();
26     vector<int> sz(n), cents;
27     function<void(int, int)> dfs1 = [&](int u, int p) {
28         sz[u] = 1;
29         bool can = 1;
30         for (int v : t[u]) if (v ^ p) {
31             dfs1(v, u);
32             if (sz[v] * 2 > n) can = 0;
33             sz[u] += sz[v];
34         }
35         if (n - sz[u] > n/2) can = 0;
36         if (can) cents.push_back(u);
37     };
38     dfs1(0, 0);
39     return min(treehash(cents.front(), -1, t), treehash(cents.back(), -1, t));
40 }

```

Binary Lifting

```

1  int Log = __lg(n) + 1;
2  vector<vector<int>> lift(n+1, vector<int>(Log, -1));
3  function<void(int, int)> dfs2 = [&](int u, int p) {
4      for (auto v : tree[u]) if (v != p) {
5          lift[v][0] = u;
6          for (int l = 1; l < Log && ~lift[v][l-1]; l++)
7              lift[v][l] = lift[ lift[v][l-1] ][l-1];
8          dfs2(v, u);
9      }
10 };
11 dfs2(root, -1);
12
13 if (depth[u] < depth[v])
14     swap(u, v);
15 int k = depth[u] - depth[v];
16
17 for (int l = Log-1; ~l && ~u; l--) {
18     if (k >> l & 1)
19         u = lift[u][l];
20 }
21 if (u == v) return u;
22 for (int l = Log-1; l > -1; l--) {
23     if (lift[u][l] != lift[v][l])
24         u = lift[u][l], v = lift[v][l];
25 }
26 return lift[u][0];

```

LCA

```

1  struct LCA {

```

```

2   int n, m, LOG;
3   vector<vector<int>> g;
4   vector<int> euler, first, depth, lg;
5   vector<vector<int>> st;
6
7   LCA(const vector<vector<int>>& tree, int root) {
8       n = tree.size();
9       g = tree;
10      first.assign(n, -1);
11      depth.assign(n, 0);
12
13      dfs(root, -1, 0);
14      m = euler.size();
15
16      lg.assign(m+1, 0);
17      for (int i = 2; i <= m; i++)
18          lg[i] = lg[i>>1] + 1;
19      LOG = lg[m] + 1;
20
21      st.assign(m, vector<int>(LOG));
22      for (int i = 0; i < m; i++)
23          st[i][0] = euler[i];
24
25      for (int j = 1; j < LOG; j++) {
26          for (int i = 0; i + (1<<j) <= m; i++) {
27              int x = st[i][j-1], y = st[i + (1<<(j-1))][j-1];
28              st[i][j] = (depth[x] < depth[y] ? x : y);
29          }
30      }
31  }
32
33  void dfs(int u, int p, int d) {
34      if (first[u] == -1) first[u] = euler.size();
35      euler.push_back(u);
36      depth[u] = d;
37      for (int v : g[u]) {
38          if (v == p) continue;
39          dfs(v, u, d+1);
40          euler.push_back(u);
41      }
42  }
43
44  int lca(int u, int v) {
45      int L = first[u], R = first[v];
46      if (L > R) swap(L, R);
47      int len = R - L + 1, k = lg[len];
48      int x = st[L][k], y = st[R - (1<<k) + 1][k];
49      return depth[x] < depth[y] ? x : y;
50  }
51
52  int dis(int u, int v) {
53      return depth[u] + depth[v] - 2 * depth[lca(u, v)];
54  }
55  };

```

DSU on Trees

```
1 void pre(int u) {
2     sz[u] = 1;
3     for (int &v : tree[u]) {
4         tree[v].erase(find(tree[v].begin(), tree[v].end(), u));
5         pre(v);
6         sz[u] += sz[v];
7         if (sz[v] > sz[tree[u].front()]) swap(v, tree[u].front());
8     }
9 }
10
11 void addver(int u) {
12 }
13
14 void removever(int u) {
15 }
16
17
18 void addsub(int u) {
19     addver(u);
20     for (int v : tree[u]) addsub(v);
21 }
22
23 void removesub(int u) {
24     removever(u);
25     for (int v : tree[u]) removesub(v);
26 }
27
28 void dfs(int u) {
29     for (int v : tree[u]) if (v != tree[u].front()) {
30         dfs(v);
31         removesub(v);
32     }
33     if (!tree[u].empty())
34         dfs(tree[u].front());
35     addver(u);
36     for (int v : tree[u])
37         if (v != tree[u].front())
38             addsub(v);
39     // query[u]
40 }
```

SCC / Strongly Connected Components

```
1 // --- USAGE REQUIREMENTS ---
2 // 1. timer must be declared and initialized to 0 outside the function.
3 // 2. g is your 0-based directed adjacency list (vector<vector<int>>).
4 // 3. v is a 0-based array/vector of vertex weights
5     (used to find the min weight per SCC).
6 // 4. idsccl[i] will hold the 0-based SCC ID for vertex i.
7 // 5. cond generates the condensed DAG where each node is a distinct SCC.
8
9 // --- MODIFICATIONS FOR BRIDGES & ARTICULATION POINTS (UNDIRECTED GRAPH) ---
10 // 1. Update the lambda signature to pass the parent: tarj = [&](int u, int p = -1)
11 // 2. Add int children = 0; at the top of the lambda to
```

```

12     count root children for APs.
13 // 3. Ignore the back-edge to the parent in the loop: if (v_ == p) continue;
14 // 4. You can completely remove vis, stk, mn, and idsccl arrays if you are not extracting SCCs.
15
16
17 vector<int> tin(n, -1), low(n), vis(n), stk, mn, idsccl(n, -1);
18 function<void(int)> tarj = [&](int u) { // Change to: (int u, int p = -1)
19     tin[u] = low[u] = timer++;
20     vis[u] = 1; // Marks u as currently in the active SCC stack
21     stk.push_back(u);
22
23     // int children = 0; // Uncomment for Articulation Points
24
25     for (int v_ : g[u]) {
26         // Uncomment to prevent traversing back to parent in undirected graphs
27         // if (v_ == p) continue;
28
29         if (tin[v_] == -1) {
30             // children++; // Uncomment for Articulation Points
31             tarj(v_); // Change to: tarj(v_, u)
32             low[u] = min(low[u], low[v_]);
33
34             // --- BRIDGE CHECK ---
35             // if (low[v_] > tin[u]) { /* Edge (u, v_) is a bridge */ }
36
37             // --- ARTICULATION POINT CHECK (NON-ROOT) ---
38             // if (low[v_] >= tin[u] && p != -1) { /* Vertex u is an AP */ }
39
40         } else if (vis[v_]) {
41             // Back-edge found. v_ is already visited AND
42             // still in the current stack
43             low[u] = min(low[u], tin[v_]);
44         }
45     }
46
47     // --- SCC EXTRACTION LOGIC ---
48     // If u is the root of an SCC, pop all vertices belonging to
49     // it from the stack
50     if (low[u] == tin[u]) {
51         int t;
52         mn.push_back(1e5); // Initialize the minimum weight for this new SCC
53         do {
54             t = stk.back();
55             stk.pop_back();
56             vis[t] = 0; // Mark as removed from the active stack
57             // Track min value (requires your v array)
58             mn.back() = min(mn.back(), v[t]);
59             // Assign the new SCC ID to vertex t
60             idsccl[t] = (int)mn.size() - 1;
61         } while (t ^ u);
62     }
63
64     // --- ARTICULATION POINT CHECK (ROOT) ---
65     // if (p == -1 && children > 1) { /* Vertex u (the root) is an AP */ }
66 };
67
68 for (int i = 0; i < n; i++)

```

```

69     if (tin[i] == -1)
70         tarj(i); // Change to: tarj(i, -1)
71
72 // --- BUILD THE CONDENSED DAG ---
73 vector<vector<int>> cond(mn.size());
74 for (int i = 0; i < n; i++)
75     for (int u : g[i])
76         if (idscc[i] ^ idscc[u]) // If an edge connects two different SCCs
77             cond[idscc[i]].push_back(idscc[u]);

```

WLCA

```

1  template<class T>
2  struct WLCA {
3      int n, Log;
4      vector<vector<int>> up;
5      vector<vector<T>> val;
6      vector<int> lvl;
7      explicit WLCA(vector<vector<pair<int, T>>> &g, int root = 0) : n((int)g.size()),
8          lvl(n), Log(__lg(n) + 1), up(n, vector<int>(Log, root)),
9          val(n, vector<T>(Log)) {
10         function<void(int)> dfs = [&](int u) -> void {
11             for(auto [v, w] : g[u]) {
12                 if(v == up[u][0]) continue;
13                 lvl[v] = lvl[u] + 1, up[v][0] = u, val[v][0] = w;
14                 for(int l = 1; l < Log; l++) {
15                     up[v][l] = up[up[v][l - 1]][l - 1];
16                     val[v][l] = val[v][l - 1] + val[up[v][l - 1]][l - 1];
17                 }
18                 dfs(v);
19             }
20         };
21         dfs(root);
22     }
23     pair<int, T> k_ancestor(int u, int k) {
24         T ans;
25         while(k) {
26             ans = ans + val[u][__builtin_ctz(k)];
27             u = up[u][__builtin_ctz(k)];
28             k &= k - 1;
29         }
30         return {u, ans};
31     }
32     int lca(int u, int v) {
33         if(lvl[u] < lvl[v]) swap(u, v);
34         u = k_ancestor(u, lvl[u] - lvl[v]).first;
35         if(u == v) return u;
36         for(int l = Log - 1; l >= 0; l--) {
37             if(up[u][l] ^ up[v][l]) {
38                 u = up[u][l], v = up[v][l];
39             }
40         }
41         return up[u][0];
42     }
43     int dis(int u, int v, int l = -1) {
44         if(l == -1) l = lca(u, v);

```



```

45     return lvl[u] + lvl[v] - 2 * lvl[l];
46 }
47 };

```

Rerooter

```

1 namespace reroot {
2     const auto exclusive = [] (const auto& a, const auto& base,
3                               const auto& merge_into, int vertex) {
4         int n = (int)a.size();
5         using Aggregate = decay_t<decltype(base)>;
6         vector<Aggregate> b(n, base);
7         for (int bit = __lg(n); bit >= 0; --bit) {
8             for (int i = n - 1; i >= 0; --i) b[i] = b[i >> 1];
9             int sz = n - (n & !bit);
10            for (int i = 0; i < sz; ++i) {
11                int index = (i >> bit) ^ 1;
12                b[index] = merge_into(b[index], a[i], vertex, i);
13            }
14        }
15        return b;
16    };
17    // MergeInto : Aggregate * Value * Vertex(int) * EdgeIndex(int) -> Aggregate
18    // Base : Vertex(int) -> Aggregate
19    // FinalizeMerge : Aggregate * Vertex(int) * EdgeIndex(int) -> Value
20    const auto rerooter = [] (const auto& g, const auto& base,
21                              const auto& merge_into, const auto& finalize_merge) {
22        int n = (int)g.size();
23        using Aggregate = decay_t<decltype(base(0))>;
24        using Value = decay_t<decltype(finish_merge(base(0), 0, 0))>;
25        vector<Value> root_dp(n), dp(n);
26        vector<vector<Value>> edge_dp(n), redge_dp(n);
27
28        vector<int> bfs, parent(n);
29        bfs.reserve(n);
30        bfs.push_back(0);
31        for (int i = 0; i < n; ++i) {
32            int u = bfs[i];
33            for (auto v : g[u]) {
34                if (parent[u] == v) continue;
35                parent[v] = u;
36                bfs.push_back(v);
37            }
38        }
39
40        for (int i = n - 1; i >= 0; --i) {
41            int u = bfs[i];
42            int p_edge_index = -1;
43            Aggregate aggregate = base(u);
44            for (int edge_index = 0; edge_index < (int)g[u].size(); ++edge_index) {
45                int v = g[u][edge_index];
46                if (parent[u] == v) {
47                    p_edge_index = edge_index;
48                    continue;
49                }

```

```

50         aggregate = merge_into(aggregate, dp[v], u, edge_index);
51     }
52     dp[u] = finalize_merge(aggregate, u, p_edge_index);
53 }
54
55 for (auto u : bfs) {
56     dp[parent[u]] = dp[u];
57     edge_dp[u].reserve(g[u].size());
58     for (auto v : g[u]) edge_dp[u].push_back(dp[v]);
59     auto dp_exclusive = exclusive(edge_dp[u], base(u), merge_into, u);
60     redge_dp[u].reserve(g[u].size());
61     for (int i = 0; i < (int)dp_exclusive.size(); ++i)
62         redge_dp[u].push_back(finalize_merge(dp_exclusive[i], u, i));
63     root_dp[u] = finalize_merge(n > 1 ?
64         merge_into(dp_exclusive[0], edge_dp[u][0], u, 0) :
65         base(u), u, -1);
66     for (int i = 0; i < (int)g[u].size(); ++i) {
67         dp[g[u][i]] = redge_dp[u][i];
68     }
69 }
70
71 return make_tuple(move(root_dp), move(edge_dp), move(redge_dp));
72 };
73 } // namespace reroot
74 // [&](Aggregate agg, Aggregate chdp, int v, int eid)
75 // [&](Aggregate agg, int v, int eid);

```

Dynamic Connectivity

```

1 struct Query {
2     char t;
3     int u, v;
4 };
5
6 struct Elem {
7     int u, v, szU, cnt;
8 };
9
10 struct DSURollback {
11     // offline : [+ a b] add edge between a, b
12     //           [- a b] remove edge between a, b
13     //           [?] number of connected components
14     int cnt;
15     stack<Elem> st;
16     vector<bool> ans;
17     vector<int> sz, par;
18     map<int, vector<pair<int, int>>> g;
19
20     DSURollback(int n) {
21         cnt = n;
22         par.resize(n + 1);
23         sz.resize(n + 1, 1);
24         iota(par.begin(), par.end(), 0);
25     }
26
27     void rollback(int x) {

```

```

28     while (st.size() > x) {
29         auto e = st.top();
30         st.pop();
31         cnt = e.cnt;
32         sz[e.u] = e.szU;
33         par[e.v] = e.v;
34     }
35 }
36
37 int findSet(int u) {
38     return par[u] == u ? u : findSet(par[u]);
39 }
40
41 void update(int u, int v) {
42     st.push({u, v, sz[u], cnt});
43     cnt--;
44     par[v] = u;
45     sz[u] += sz[v];
46 }
47
48 void unionSet(int u, int v) {
49     u = findSet(u);
50     v = findSet(v);
51     if (u != v) {
52         if (sz[u] < sz[v])
53             swap(u, v);
54         update(u, v);
55     }
56 }
57
58 void solve(int x, int l, int r) {
59     int cur = st.size();
60
61     for (auto i: g[x])
62         unionSet(i.first, i.second);
63
64     if (l == r) {
65         if (ans[l])
66             cout << cnt << endl;
67         rollback(cur);
68         return;
69     }
70     int m = (l + r) >> 1;
71     solve(x * 2, l, m);
72     solve(x * 2 + 1, m + 1, r);
73     rollback(cur);
74 }
75
76 void traverse(int x, int lX, int rX, int l, int r, int u, int v) {
77     if (rX < l || lX > r)
78         return;
79     if (lX >= l && rX <= r) {
80         g[x].emplace_back(u, v);
81         return;
82     }
83     int m = (lX + rX) >> 1;

```

```

84         traverse(x * 2, lX, m, l, r, u, v);
85         traverse(x * 2 + 1, m + 1, rX, l, r, u, v);
86     }
87
88     void build(vector<Query> &queries) {
89         int q = queries.size();
90         ans.resize(q);
91         map<pair<int, int>, vector<pair<int, int>>> mp;
92         for (int i = 0; i < queries.size(); i++) {
93             auto cur = queries[i];
94             if (cur.u > cur.v)
95                 swap(cur.u, cur.v);
96             if (cur.t == '?')
97                 ans[i] = 1;
98             else if (cur.t == '+')
99                 mp[{cur.u, cur.v}].emplace_back(i, queries.size());
100             else {
101                 mp[{cur.u, cur.v}].back().second = i - 1;
102                 traverse(1, 0, q - 1, mp[{cur.u, cur.v}].back().first,
103                     mp[{cur.u, cur.v}].back().second, cur.u, cur.v);
104             }
105         }
106
107         for (auto i: mp) {
108             for (auto j: i.second) {
109                 if (j.second == q)
110                     traverse(1, 0, q - 1, j.first, q - 1, i.first.first,
111                         i.first.second);
112             }
113         }
114     };
115
116     void testCase() {
117         int n, q;
118         cin >> n >> q;
119         if (!q)
120             return;
121         DSURollback dsu(n);
122         vector<Query> queries(q);
123         char t;
124         int x, y;
125         for (int i = 0; i < q; i++) {
126             cin >> t;
127             if (t == '?')
128                 queries[i] = {t, 0, 0};
129             else {
130                 cin >> x >> y;
131                 if (y < x)
132                     swap(x, y);
133                 queries[i] = {t, x, y};
134             }
135         }
136
137         dsu.build(queries);
138         dsu.solve(1, 0, q - 1);
139

```

140 }

Max Flow (Dinic)

```
1  class Dinic { // O(n^2 * m), 0-based
2      private:
3          struct E {
4              int to, rev;
5              ll c, oc;
6              ll f() { return max(oc - c, 0LL); }
7          };
8          vector<int> lvl, ptr, q;
9          vector<vector<E>> g;
10         ll dfs(int v, int t, ll f) {
11             if (v == t || !f) return f;
12             for (int &i = ptr[v]; i < g[v].size(); ++i) {
13                 E &e = g[v][i];
14                 if (lvl[e.to] == lvl[v] + 1 && e.c)
15                     if (ll p = dfs(e.to, t, min(f, e.c))) {
16                         e.c -= p;
17                         g[e.to][e.rev].c += p;
18                         return p;
19                     }
20             }
21             return 0;
22         }
23
24     public:
25         Dinic(int n) : lvl(n), ptr(n), q(n), g(n) {}
26         void add_edge(int u, int v, ll c) { // directed
27             g[u].push_back({v, (int)g[v].size(), c, c});
28             g[v].push_back({u, (int)g[u].size() - 1, 0, 0});
29         }
30         ll calc(int s, int t) { // source to destination
31             ll flow = 0;
32             while (true) {
33                 fill(lvl.begin(), lvl.end(), 0);
34                 int qs = 0, qe = 0;
35                 lvl[q[qe++] = s] = 1;
36                 while (qs < qe && !lvl[t]) {
37                     int v = q[qs++];
38                     for (auto &e : g[v])
39                         if (!lvl[e.to] && e.c) lvl[q[qe++] = e.to] = lvl[v] + 1;
40                 }
41                 if (!lvl[t]) break;
42                 fill(ptr.begin(), ptr.end(), 0);
43                 while (ll p = dfs(s, t, LLONG_MAX)) flow += p;
44             }
45             return flow;
46         }
47         void reset() { // before calling calc again
48             for (auto &adj : g)
49                 for (auto &e : adj) e.c = e.oc;
50         }
51         bool leftOfMinCut(int a) { return lvl[a] != 0; }
52     };
```

Max Bipartite Matching (Karp) [with building]

```
1 struct matching {
2     int nL, nr;
3     vector<vector<int>> g;
4     vector<int> dis, mL, mr;
5     explicit matching(int nL, int nr) : nL(nL), nr(nr), g(nL), dis(nL), mL(nL, -1),
6     mr(nr, -1) { }
7
8     void add(int l, int r) { // [i, j]
9         g[l].push_back(r);
10    }
11
12    void bfs() {
13        queue<int> q;
14        for(int u = 0; u < nL; u++) {
15            if(mL[u] == -1) q.push(u), dis[u] = 0;
16            else dis[u] = -1;
17        }
18        while(!q.empty()) {
19            int l = q.front(); q.pop();
20            for(int r : g[l]) {
21                if(mr[r] != -1 && dis[mr[r]] == -1)
22                    q.push(mr[r]), dis[mr[r]] = dis[l] + 1;
23            }
24        }
25    }
26
27    bool canMatch(int l) {
28        for(int r : g[l]) if(mr[r] == -1)
29            return mr[r] = l, mL[l] = r, true;
30        for(int r : g[l]) if(dis[l] + 1 == dis[mr[r]] && canMatch(mr[r]))
31            return mr[r] = l, mL[l] = r, true;
32        return false;
33    }
34
35    int maxMatch() {
36        int ans = 0, turn = 1;
37        while(turn) {
38            bfs(), turn = 0;
39            for(int l = 0; l < nL; l++) if(mL[l] == -1)
40                turn += canMatch(l);
41            ans += turn;
42        }
43        return ans;
44    }
45
46    pair<vector<int>, vector<int>> minCover() {
47        vector<int> L, R;
48        for (int u = 0; u < nL; ++u) {
49            if(dis[u] == -1) L.push_back(u);
50            else if(mL[u] != -1) R.push_back(mL[u]);
51        }
52        return {L, R};
53    }
54 };
```

4. Math and Number Theory

Math

```
1 Stirling numbers of the first kind : the number of permutations of n elements with k disjoint
2  $S(n,k) = n * S(n-1,k) + S(n-1,k-1)$ 
3 Sum  $k = 0 \rightarrow n$  of  $S(n,k) = n!$ 
4
5 Stirling numbers of the second kind : the number of ways to partition
6 a set of n elements into k nonempty subsets
7  $S(n,k) = k * S(n-1,k) + S(n-1,k-1)$ 
8 Sum  $k = 0 \rightarrow n$  of  $S(n,k) = B_n$ 
9
10 Bell numbers : the possible partitions of a set into nonempty subsets
11 Sum  $k = 0 \rightarrow n-1$  of  $n-1Ck * B_k = B_n$ 
12
13 Sum of first n even numbers :  $n*(n+1)$ 
14 Sum of first n odd numbers :  $n*n$ 
15
16 Sum of squares of first n numbers:
17  $n*(n+1)*(2*n+1)/6$ 
18 Sum of squares of first n even numbers:
19  $2*n*(n+1)*(2*n+1)/3$ 
20 Sum of squares of first n odd numbers:
21  $n*(2*n+1)*(2*n-1)/3$ 
22
23 Number of ways to pick equal number of elements from two sets :  $(n+m)C(m)$ 
24
25 Sum of  $\phi(d)$  for all  $d | n$  is equal to n.
26 Number of pairs (x,y) that satisfy  $x+y=n$  and  $\gcd(x,y)=1$  is  $\phi(n)$ .
27
28 Sum  $(nCk)$  for  $k [0,n] = 2^n$ 
29 Sum  $(mCk)$  for all  $m [0,n] = (n+1)C(k+1)$ 
30 Sum  $((n+k)Ck)$  for all  $k [0,m] = (n+m+1)C(m)$ 
31 Sum  $((nCk)^2)$  for all  $k [0,n] = (2*n)Cn$ 
32 Sum  $(i*nCi)$  for all  $i [1,n] = n*2^{(n-1)}$ 
33 Sum  $((n-i)Ci)$  for all  $i [0,n] = F(n+1)$ 
34 Number of arrays with size n and sum m =  $(n-1+m)C(m) = (n-1+m)C(n-1)$ 
35
36  $P(A|B)$  is the probability of event A given that event B happened.
37  $P(A\&B)$  is the probability of events A and B happening.
38
39  $P(A|B) = (P(B|A) * P(A)) / P(B)$ 
40  $P(A|B) = P(A\&B) / P(B)$ 
41
42 Divisibility:
43 2 if the rightmost digit is divisible by 2
44 3 if the sum of the digits is divisible by 3
45 4 if the number formed by the last two digits is divisible by 4
46 5 if the rightmost digit is 0 or 5
47 6 if it is divisible by 2 and 3
48 7 if The number formed by all digits except the right-most digit -
49 (2 * right-most digit) is divisible by 7
50 8 if the number formed by the last 3 digits is divisible by 8
51 9 if the sum of the digits is divisible by 9
52
53 -Getting half of the binomial expansion (Only odd indices or only even indices)
```

```

54     by using  $(a+b)^n$  and  $(a-b)^n$  and adding both of them
55
56 To solve quadratic equation  $ax^2 + bx + c = 0$ 
57  $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ 
58
59 -Sum of geometric series  $ar^i$  (i from 0 to n) =  $a(1 - r^{n+1}) / (1 - r)$ 
60 -Sum of geometric series  $ar^i$  (i from 0 to infinity) =  $a / (1 - r)$ 
61
62 - XOR from 1 to n =  $n \% 4$  [n, 1, n+1, 0]
63
64 - n can be sum of 2 squares if n has no  $p^k$  [p = 3 % 4, k is odd]
65     in its prime factorization
66 - n can be sum of 3 squares if n is not in form  $4^a(8b + 7)$ 
67 - n always can be sum of  $\geq 4$  squares (remaining are zeros)
68
69 ===== FADY THE GOOOOAAAT =====
70     sum of divisors
71     primepower * prime2power2 * ...
72
73     ((prime(power + 1) - 1) / (prime - 1)) * ((prime2(power2 + 1) - 1) / (prime2 - 1)) * ...
74 =====
75     a % m == b
76     a and m not coprime
77     g = gcd(a, m)
78     (a / g) % (m / g) = b / g
79
80     ax % m == b
81     a and m not coprime
82     g = gcd(a, m)
83     (a(x-1) * (a / g)) % (m / g) = b / g
84 =====
85     a(power%phi(m)) % m;
86 =====
87     count balanced brackets
88     r=n/2 || or r = number of opened brackets
89     nCr(n, r) - nCr(n, r-1)
90 =====
91     // different n*n grids whose each square have m colors
92     // if possible to rotate one of them so that they look the same then they same
93     t = n * n;
94     total = (fp(m, t)
95         + fp(m, (t + 1) / 2)
96         + 2 * fp(m, (t / 4) + (n % 2))) % mod;
97     total = mul(total, fp(4, mod - 2));
98 =====
99     biggest divisors
100    735134400 1344 => 26 33 52 7 11 13 17
101    73513440 768
102 =====
103    for (int x = mask; x > 0; x = (x - 1) & mask)
104        get all x such that mask = mask | x
105 =====
106    sum from 1 to n: i * nCr(n, i) = n * (1LL << (n - 1))
107    sum of odd between [1, n] = ((n+1) / 2)2
108    sum of even between [1, n] = (n/2) * (n/2 + 1)
109
110 ll sum_total(ll l, ll r) { // sum [l, r]

```



```

111     return (r - l + 1) * (l + r) / 2;
112 }
113
114 ll sum_even(ll l, ll r) { // sum even [l, r]
115     if (l % 2 != 0) ++l;
116     if (r % 2 != 0) --r;
117     if (l > r) return 0;
118     ll n = (r - l) / 2 + 1;
119     return n * (l + r) / 2;
120 }
121
122 ll sum_odd(ll l, ll r) { // sum odd [l, r]
123     if (l % 2 == 0) ++l;
124     if (r % 2 == 0) --r;
125     if (l > r) return 0;
126     ll n = (r - l) / 2 + 1;
127     return n * (l + r) / 2;
128 }
129
130 ll arithm1(ll l, ll r, ll a, ll d) { // [l, r] starting a and diff d
131     if (d == 0) return (a >= l && a <= r) ? a : 0;
132     ll n1 = ((l - a + d - (d > 0 ? 1 : -1)) / d);
133     ll first = a + n1 * d;
134     if ((d > 0 && first > r)
135         || (d < 0 && first < r))
136         return 0;
137     ll n2 = ((r - a) / d);
138     ll last = a + n2 * d;
139     ll n = (n2 - n1 + 1);
140     return n * (first + last) / 2;
141 }
142
143 ll arithm2(ll a, ll d, ll n) { // starting a, diff d, n terms
144     return n * (2 * a + (n - 1) * d) / 2;
145 }
146 */
147
148 ll phi(ll x) { // sqrt(x)
149     ll ans = x;
150     for(ll i = 2; i * i <= x; i++) {
151         if(x % i == 0) {
152             while(x % i == 0) x /= i;
153             ans -= ans / i;
154         }
155     }
156     if(x > 1) ans -= ans / x;
157     return ans;
158 }
159
160 array<ll, 2> CRT(ll a1, ll m1, ll a2, ll m2) {
161     // x = a1 % m1, x = a2 % m2
162     a1 %= m1, a2 %= m2;
163     auto [g, q1, q2] = eGcd(m1, -m2);
164     if ((a2 - a1) % g) return {-1, -1};
165     ll lcm = m1 / g * m2;
166     ll m = m2 / g;

```

```

167     q1 = (a2 - a1) / g % m * q1 % m;
168     ll res = (a1 + m1 * q1) % lcm;
169     if (res < 0) res += lcm;
170     return {res, lcm};
171 }
172
173 ll BSGS(ll a, ll b, ll p) { //  $a^x = b \pmod p$ 
174     a %= p, b %= p;
175     if(b == 1) return 0;
176     if(a == 0) return b == 0? 1: -1;
177     int add = 0;
178     ll g, tmp = 1;
179     while ((g = gcd(a, p)) > 1) {
180         if(b % g) return -1;
181         p /= g, b /= g, tmp = tmp * (a / g) % p, ++add;
182         if(tmp == b) return add;
183     }
184     b = b * modInv(tmp, p) % p;
185     int n = (int)sqrtl(p) + 1;
186     unordered_map<ll, int> mp;
187     for (ll q = 0, cur = 1; q <= n; ++q)
188         mp.emplace(cur, q), cur = cur * a % p;
189     ll an = 1;
190     for (ll i = 0; i < n; ++i) an = an * a % p;
191     an = modInv(an, p);
192     for (ll i = 0, cur = b; i <= n; ++i) {
193         auto it = mp.find(cur);
194         if(it != mp.end()) return i * n + it->second + add;
195         cur = cur * an % p;
196     }
197     return -1;
198 }
199
200 int sumNPowerM(int n, int m) { //  $1^m + 2^m \dots n^m$ 
201     int k = m + 3;
202     vector<int> res(k);
203     for(int i = 1; i < k; i++) res[i] = (res[i - 1] + fp(i, m)) % mod;
204     if(n < k) return res[n];
205     int facK = k;
206     vector<int> p(k); p[0] = 1;
207     for(int i = 1; i < k; i++) {
208         p[i] = int(p[i - 1] * 1LL * (n - i) % mod);
209         facK = int(facK * 1LL * i % mod);
210     }
211     vector<int> inv(k + 1), s(k + 1);
212     inv[k] = fp(facK, mod - 2), s[k] = 1;
213     for(int i = k - 1; i >= 0; i--) {
214         s[i] = int(s[i + 1] * 1LL * (n - i) % mod);
215         inv[i] = int(inv[i + 1] * 1LL * (i + 1) % mod);
216     }
217     int ans = 0;
218     for(int i = 1; i < k; i++) {
219         int cur = int(res[i] * 1LL * p[i - 1] % mod * s[i + 1] % mod *
220             inv[i - 1] % mod * inv[k - i - 1] % mod);
221         if((k - i + 1) & 1) cur = (mod - cur) % mod;
222         ans = (ans + cur) % mod;

```

```

223     }
224     return ans;
225 }

```

ModInt

```

1  template <int M> struct ModInt {
2      template <class T> T binpow(T a, ll b) {
3          T res = 1;
4          while (b) { if (b & 1) res *= a; a *= a, b >>= 1; }
5          return res;
6      }
7      int v;
8      ModInt() : v(0) {}
9      ModInt(ll v_) {
10         v = v_ % M;
11         if (v < 0) v += M;
12     }
13
14     bool operator==(ModInt o) const { return v == o.v; };
15     bool operator!=(ModInt o) const { return !(*this == o); };
16
17     // ModInt add(ModInt a, ModInt b) { return ((a.v + b.v % M)+M)%M; }
18     // ModInt sub(ModInt a, ModInt b) { return add(a, -b); }
19     // ModInt mul(ModInt a, ModInt b) { return 1ll * a.v * b.v % M; }
20     // ModInt div(ModInt a, ModInt b) { return mul(a, binpow(b, M-2)); }
21
22     ModInt &operator+=(ModInt o) { v = (v + o.v) % M; return *this; }
23     ModInt &operator-=(ModInt o) { v = (v - o.v + M) % M; return *this; }
24     ModInt &operator*=(ModInt o) { v = 1ll * v * o.v % M; return *this; }
25     ModInt &operator/=(ModInt o) { return (*this *= binpow(o, M - 2)); }
26
27     friend ModInt operator+(ModInt a, ModInt b) { return a += b; }
28     friend ModInt operator-(ModInt a, ModInt b) { return a -= b; }
29     friend ModInt operator*(ModInt a, ModInt b) { return a *= b; }
30     friend ModInt operator/(ModInt a, ModInt b) { return a /= b; }
31     ModInt operator-() { return 0 - *this; }
32
33     friend istream &operator>>(istream &is, ModInt &a) {
34         ll x; is >> x;
35         a = ModInt(x);
36         return is;
37     }
38     friend ostream &operator<<(ostream &os, ModInt a) { return os << a.v; }
39     friend string to_string(ModInt a) { return to_string(a.v); }
40 };
41 using mint = ModInt<>;

```

Sieve / PHI up to n / 2D gcd()

```

1  const int NS = 1e7;
2  const int NP = (NS/log(NS)) * (1 + 1.28 / log(NS));
3  int prSz;
4  int spf[NS], prm[NP];
5

```

```

6  auto pre_Sieve = []() {
7      for (int i = 2; i < NS; i++){
8          if(!spf[i]) spf[i] = prm[prSz++] = i;
9          for(int j = 0; i * prm[j] < NS; j++) {
10             spf[i * prm[j]] = prm[j];
11             if(spf[i] == prm[j]) break;
12         }
13     }
14     return 0;
15 }();
16
17 auto factors(int n) {
18     vector<array<int, 2>> res;
19     if(n < 2) return res;
20     int p = spf[n];
21     while(p > 1) {
22         res.push_back({p, 0});
23         while(n % p == 0) n /= p, res.back()[1]++;
24         p = spf[n];
25     }
26     return res;
27 }
28
29 auto getDivisors(int _n) {
30     auto _fac = factors(_n);
31     int cnt = 1;
32     for(auto [pr, pw] : _fac) cnt *= pw + 1;
33     vector<int> res(1, 1); res.reserve(cnt);
34
35     for(auto [pr, pw] : _fac)
36         for(int i = int(res.size()) - 1; i >= 0; i--)
37             for(int b = pr, j = 0; j < pw; j++, b *= pr)
38                 res.push_back(res[i] * b);
39     sort(res.begin(), res.end());
40     return res;
41 }
42
43 bool isPrime(ll n) {
44     if(n < 4) return n > 1;
45     if(n % 2 == 0 || n % 3 == 0) return false;
46     for (ll i = 5; i * i <= n; i += 6)
47         if (n % i == 0 || n % (i + 2) == 0)
48             return false;
49     return true;
50 }
51
52 void phi_1_to_n(int n) {
53     vector<int> phi(n + 1);
54     for (int i = 0; i <= n; i++)
55         phi[i] = i;
56
57     for (int i = 2; i <= n; i++) {
58         if (phi[i] == i) {
59             for (int j = i; j <= n; j += i)
60                 phi[j] -= phi[j] / i;
61         }
62     }

```

```

62     }
63 }
64
65 // 2d gcd in n^2
66 vector<vector<int>> GCD(N, vector<int>(N, 1));
67 for(int d = 2; d < N; ++d)
68     for(int i = d; i < N; i += d)
69         for(int j = d; j < N; j += d)
70             GC[i][j] = d;

```

Sieve up to 1e9

```

1  vector<int> sieve(const int BN, const int Q = 17, const int L = 1 << 15) {
2      using u = uint32_t; using u8 = uint8_t;
3      static const u rs[] = {1, 7, 11, 13, 17, 19, 23, 29};
4      struct P {
5          u p;
6          u pos[8];
7      };
8
9      const u v = sqrt(BN), vv = sqrt(v);
10     vector<bool> isp(v + 1, true);
11     for (u i = 2; i <= vv; ++i)
12         if (isp[i])
13             for (u j = i * i; j <= v; j += i) isp[j] = false;
14
15     const u rsize = BN > 60184 ? BN / (log(BN) - 1.1) :
16         max(1.0, BN / (log(BN) - 1.11)) + 1 + 30;
17     vector<int> primes = {2, 3, 5}; u psize = 3;
18     primes.resize(rsize); vector<P> sprimes;
19
20     u pbeg = 0, prod = 1;
21     for (u p = 7; p <= v; ++p) {
22         if (!isp[p]) continue;
23         if (p <= Q) prod *= p, ++pbeg, primes[psize++] = p;
24         P pp = {p, {}};
25         for (int t = 0; t < 8; ++t) {
26             u j = p <= Q ? p : p * p;
27             while (j % 30 != rs[t]) j += p << 1;
28             pp.pos[t] = j / 30;
29         }
30         sprimes.push_back(pp);
31     }
32
33     vector<u8> pre(prod, 0xFF);
34     for (size_t pi = 0; pi < pbeg; ++pi) {
35         auto &pp = sprimes[pi];
36         const u p = pp.p;
37         for (int t = 0; t < 8; ++t) {
38             const u8 m = ~(1 << t);
39             for (u i = pp.pos[t]; i < prod; i += p) pre[i] &= m;
40         }
41     }
42
43     const u block_size = (L + prod - 1) / prod * prod;
44     vector<u8> block(block_size);

```

```

45     u8* __restrict pblock = block.data();
46     const u M = (BN + 29) / 30;
47
48     for (u beg = 0; beg < M; beg += block_size, pblock -= block_size) {
49         u end = min(M, beg + block_size);
50
51         for (u i = beg; i < end; i += prod)
52             memcpy(pblock + i, pre.data(), min(prod, end - i));
53         if (beg == 0) pblock[0] &= 0xFE;
54
55         for (size_t pi = pbeg; pi < sprimes.size(); ++pi) {
56             auto &pp = sprimes[pi];
57             const u p = pp.p;
58             #pragma GCC unroll 8
59             for (int t = 0; t < 8; ++t) {
60                 u i = pp.pos[t];
61                 const u8 m = ~(1 << t);
62                 for (; i < end; i += p) pblock[i] &= m;
63                 pp.pos[t] = i;
64             }
65         }
66
67         for (u i = beg; i < end; ++i)
68             for (u m = pblock[i]; m > 0; m &= m - 1)
69                 primes[psize++] = i * 30 + rs[__builtin_ctz(m)];
70     }
71
72     while (psize > 0 && primes[psize - 1] > BN) --psize;
73     primes.resize(psize); return primes;
74 }

```

Egcd, Linear Diophantine

```

1  array<ll, 3> eGcd(ll a, ll b) {
2      if (b == 0) return {a, 1, 0};
3      auto [g, x1, y1] = eGcd(b, a % b);
4      return {g, y1, x1 - (a / b) * y1};
5  }
6
7  ax0 + by0 = g ==> ax + by = c
8  x0 *= c/g, y0 *= c/g
9
10 all solutions:
11     x = x0 + k * (b/g),
12     y = y0 - k * (a/g);
13
14 int ceildiv(int a, int b) {
15     if ((a ^ b) >= 0) return (a + b - 1) / b;
16     else return a / b;
17 }
18
19 int flordiv(int a, int b) {
20     if ((a ^ b) >= 0) return a / b;
21     else return (a - b + 1) / b;
22 }
23

```

```

24 non-negative solution:
25     k >= ceildiv(-x*g,b)
26     k <= flordiv(x*g, a)
27
28 bool havenonnegsol(int a, int b, int c, int& x, int& y) {
29     int g = egcd(a, b, x, y);
30     x *= c/g;
31     y *= c/g;
32     int l1 = ceildiv(-x*g, b);
33     int l2 = flordiv(y*g, a);
34     return l1 <= l2;
35 }
36
37 // MOD INV
38 ll modInv(ll a, ll m) {
39     ll x = 1, x1 = 0, q, t, b = m;
40     while(b) {
41         q = a / b;
42         a -= q * b, t = a, a = b, b = t;
43         x -= q * x1, t = x, x = x1, x1 = t;
44     }
45     assert(a == 1);
46     return (x + m) % m;
47 }

```

CRT

```

1  using T = __int128;
2  // ax + by = __gcd(a, b)
3  // returns __gcd(a, b)
4  T extended_euclid(T a, T b, T &x, T &y) {
5      T xx = y = 0;
6      T yy = x = 1;
7      while (b) {
8          T q = a / b;
9          T t = b; b = a % b; a = t;
10         t = xx; xx = x - q * xx; x = t;
11         t = yy; yy = y - q * yy; y = t;
12     }
13     return a;
14 }
15 // finds x such that x % m1 = a1, x % m2 = a2. m1 and m2 may not be coprime
16 // here, x is unique modulo m = lcm(m1, m2). returns (x, m). on failure, m = -1.
17 pair<T, T> CRT(T a1, T m1, T a2, T m2) {
18     T p, q;
19     T g = extended_euclid(m1, m2, p, q);
20     if (a1 % g != a2 % g) return make_pair(0, -1);
21     T m = m1 / g * m2;
22     p = (p % m + m) % m;
23     q = (q % m + m) % m;
24     return make_pair((p * a2 % m * (m1 / g) % m +
25                      q * a1 % m * (m2 / g) % m) % m, m);
26 }

```

MillerRabin

```

1  ll binpower(ll base, ll e, ll mod) {
2      ll result = 1;
3      base %= mod;
4      while (e) {
5          if (e & 1)
6              result = (__uint128_t)result * base % mod;
7              base = (__uint128_t)base * base % mod;
8              e >>= 1;
9      }
10     return result;
11 }
12
13 bool check_composite(ll n, ll a, ll d, int s) {
14     ll x = binpower(a, d, n);
15     if (x == 1 || x == n - 1)
16         return false;
17     for (int r = 1; r < s; r++) {
18         x = (__uint128_t)x * x % n;
19         if (x == n - 1)
20             return false;
21     }
22     return true;
23 };
24
25 bool MillerRabin(long long n) {
26     if (n < 2)
27         return false;
28
29     vector<int> primes = {2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37};
30     for (auto a : primes)
31         if (n % a == 0)
32             return false;
33
34     int r = 0;
35     long long d = n - 1;
36     while ((d & 1) == 0) {
37         d >>= 1;
38         r++;
39     }
40
41     for (int a : primes) {
42         if (n == a)
43             return true;
44         if (check_composite(n, a, d, r))
45             return false;
46     }
47     return true;
48 }

```

Number of Divisors up to 1e18

```

1  namespace countdivisors {
2      // for one second: 5000 1e18, 100000 1e9, 200000 1e6
3      using ull = unsigned long long;
4      using u128 = __uint128_t;
5      static mt19937_64 rnd(chrono::steady_clock::now().time_since_epoch().count());
6      u128 mul128(ull a, ull b, ull m) { return (u128)a * b % m; }

```



```

7   ull mul_mod(ull a, ull b, ull m) { return (ull)mul128(a, b, m); }
8   ull pow_mod(ull a, ull d, ull m) {
9       ull r = 1;
10      while (d) {
11          if (d & 1) r = mul_mod(r, a, m);
12          a = mul_mod(a, a, m);
13          d >>= 1;
14      }
15      return r;
16  }
17  bool isPrime(ull n) {
18      if (n < 2) return false;
19      for (ull p : vector<ull>({2ULL,3,5,7,11,13,17,19,23,29,31,37}))
20          if (n % p == 0) return n == p;
21      ull d = n - 1, s = 0;
22      while (!(d & 1)) d >>= 1, ++s;
23      for (ull a : vector<ull>({2ULL,325,9375,28178,450775,9780504,1795265022})) {
24          if (a % n == 0) continue;
25          ull x = pow_mod(a, d, n);
26          if (x == 1 || x == n - 1) continue;
27          bool comp = true;
28          for (ull r = 1; r < s; ++r) {
29              x = mul_mod(x, x, n);
30              if (x == n - 1) { comp = false; break; }
31          }
32          if (comp) return false;
33      }
34      return true;
35  }
36  ull pollards_rho(ull n) {
37      if (n % 2 == 0) return 2;
38      while (true) {
39          ull c = uniform_int_distribution<ull>(1, n - 1)(rnd);
40          auto f = [&](ull x){ return (mul_mod(x, x, n) + c) % n; };
41          ull x = rnd() % n, y = x, d = 1;
42          while (d == 1) {
43              x = f(x); y = f(f(y));
44              d = gcd<ull>(x > y ? x - y : y - x, n);
45          }
46          if (d < n) return d;
47      }
48  }
49  void factor(ull n, map<ull,int>& cnt) {
50      if (n < 2) return;
51      if (isPrime(n)) { cnt[n]++; return; }
52      ull d = pollards_rho(n);
53      factor(d, cnt);
54      factor(n/d, cnt);
55  }
56  ull ans(ull n) {
57      map<ull,int> cnt;
58      factor(n, cnt);
59      ull res = 1;
60      for (auto [p, e] : cnt) res *= (e + 1);
61      return res;
62  }

```

```
63 }
```

n mod 1 + n mod 2 + n mod 3 + + n mod m

```
1 // n and m tends to(1e13) => time complexity(sqrt(n))
2 void solve(int idx){
3     int k=n/idx;
4     int st=n%idx%mod;
5     int l=n/(k+1);
6     int len=(idx-l)%mod;
7     ans=(ans + st*len%mod + k%mod*(len*(len-1)/2%mod)%mod)%mod;
8     if(l) solve(l);
9 }
```

n-th Fib Number

```
1 int fast_Fibonacci(int n) {
2     int i = 1, h = 1, j = 0, k = 0, t;
3     while (n > 0) {
4         if (n % 2 == 1)
5             t = j * h, j = i * h + j * k + t, i = i * k + t;
6         t = h * h, h = 2 * k * h + t, k = k * k + t, n = n / 2;
7     }
8     return j;
9 }
```

Long Division

```
1 int a = 23, b = 5, n = 10;
2 vector<int> s;
3 for (int i = 0; i < n; i++) {
4     unsigned long long k = a / b; // Integer division: gets the next digit
5     a -= b * k;                  // Remove the part already represented
6     a *= 10;                     // Shift remainder left (for next digit)
7     s.push_back(k);              // Store the digit
8 }
```

Floor Values

```
1 // code to get all different values of floor(n/i)
2 for (ll l = 1, r = 1; (n/l); l = r + 1) { // O(sqrt)
3     r = (n/(n/l));
4     // q = (n/l), process the range [l, r]
5 }
```

Combinatorics

```
1 namespace combinatorics {
2     vector<mint> fac_(1, 1), inv_(1, 1);
3     void build(int N) {
4         N += 10;
5         fac_.assign(N, 1);
6         inv_.assign(N, 1);
```

```

7      for (int i = 2; i < N; i++)
8          fac_[i] = fac_[i-1] * i;
9      inv_[N-1] = 1/fac_[N-1];
10     for (int i = N-2; i > 1; i--)
11         inv_[i] = inv_[i+1] * (i+1);
12 }
13 inline mint nCr(int n, int r) {
14     if(n < 0 || r < 0 || r > n) return 0;
15     return fac_[n] * inv_[r] * inv_[n - r];
16 }
17
18 inline mint nCr(int64_t n, int64_t r) {
19     if(n < 0 || r < 0 || r > n) return 0;
20     r = min<int64_t>(r, n - r);
21     mint ans = inv_[r];
22     for(int64_t i = n - r + 1; i <= n; i++) ans = ans * i;
23     return ans;
24 }
25 inline mint nPr(int n, int r) {
26     if(n < 0 || r < 0 || r > n) return 0;
27     return fac_[n] * inv_[n - r];
28 }
29 inline mint stars_andBars(int n, int r){
30     return nCr(n + r - 1, r - 1);
31 }
32 inline mint catalan(int n) {
33     if(n < 0) return 0;
34     return fac_[2 * n] * inv_[n] * inv_[n + 1];
35 }
36 }
37 // using namespace combinatorics;
38 auto pre = []() { combinatorics::build(); return 0; }();

```

nCr, nPr without precomputation

```

1  ll nCr(ll n, ll r) {
2      if (r < 0 || r > n) return 0;
3      r = min(r, n-r);
4      ll ret = 1;
5      for (int i = 0; i < r; i++)
6          ret = ret * (n-i) / (i+1);
7      return ret;
8  }
9  ll nPr(int n, int r) {
10     if (r < 0 || r > n) return 0;
11     ll ret = 1;
12     for (int i = 0; i < r; ++i) {
13         ret *= (n - i);
14     }
15     return ret;
16 }

```

NCR table

```

1  void preprocess_nCr() {

```

```

2     for (int n = 0; n < N; n++) {
3         C[n][0] = C[n][n] = 1;
4         for (int r = 1; r < n; r++) {
5             C[n][r] = (C[n - 1][r - 1] + C[n - 1][r]) % MOD;
6         }
7     }
8 }

```

Catalan numbers

```

1  const int MOD = ....
2  const int MAX = ....
3  int catalan[MAX];
4  void init() {
5      catalan[0] = catalan[1] = 1;
6      for (int i=2; i<=n; i++) {
7          catalan[i] = 0;
8          for (int j=0; j < i; j++) {
9              catalan[i] += (catalan[j] * catalan[i-j-1])% MOD;
10             if (catalan[i] >= MOD)
11                 catalan[i] -= MOD;
12         }
13     }
14 }
15
16 // 1- Number of correct bracket sequence consisting of n opening and n
17 // closing brackets.
18
19 // 2- The number of rooted full binary trees with n+1 leaves (vertices
20 // are not numbered).
21 // A rooted binary tree is full if every vertex has
22 // either two children or no children.
23
24 // 3- The number of ways to completely parenthesize n+1 factors.
25
26 // 4- The number of triangulations of a convex polygon with n+2 sides
27 // (i.e. the number of partitions of polygon into disjoint triangles by
28 // using the diagonals).
29
30 // 5- The number of ways to connect the 2n points on a circle to form n
31 // disjoint chords.
32
33 // 6- The number of non-isomorphic full binary trees with n internal
34 // nodes (i.e. nodes having at least one son).
35
36 // 7- The number of monotonic lattice paths from point (0,0) to point
37 // (n,n) in a square lattice of size nxn,
38 // which do not pass above the main diagonal (i.e. connecting (0,0) to
39 // (n,n)).
40
41 // 8- Number of permutations of length n that can be stack sorted
42 // (i.e. it can be shown that the rearrangement is stack sorted if and
43 // only if there is no such index i<j<k, such that ak<ai<aj).
44
45 // 9- The number of non-crossing partitions of a set of n elements.
46
47 // 10- The number of ways to cover the ladder 1..n using n rectangles

```

Matrix Exponentiation

```
1  template <typename T>
2  struct Matrix {
3      int n, m;
4      vector<T> mat;
5
6      Matrix(int n, int m) : n(n), m(m) {
7          mat.assign(n * m, 0);
8      }
9      Matrix(int n) : n(n), m(n) {
10         mat.assign(n * n, 0);
11     }
12
13     void identity() {
14         assert(n == m);
15         fill(mat.begin(), mat.end(), 0);
16         for (int i = 0; i < n; i++) {
17             mat[i * m + i] = 1;
18         }
19     }
20
21     T* operator[](int r) { return &mat[r * m]; }
22     const T* operator[](int r) const { return &mat[r * m]; }
23
24     Matrix operator*(const Matrix& o) const {
25         assert(m == o.n);
26         Matrix res(n, o.m);
27
28         for (int i = 0; i < n; i++) {
29             for (int k = 0; k < m; k++) {
30                 T val = mat[i * m + k]; if (val == 0) continue;
31                 for (int j = 0; j < o.m; j++) {
32                     res[i][j] = (res[i][j] + 1ll * val * o[k][j]) % mod;
33                     // res[i][j] = res[i][j] + val * o[k][j];
34                 }
35             }
36         }
37         return res;
38     }
39
40     Matrix operator+(const Matrix& o) const {
41         assert(o.n == n && o.m == m);
42         Matrix ret(n, m);
43         for (int i = 0; i < n; i++)
44             for (int j = 0; j < m; j++) {
45                 ret[i][j] = ((*this)[i][j] + o[i][j]) % mod;
46                 // ret[i][j] = (*this)[i][j] + o[i][j];
47             }
48         return ret;
49     }
50
51     Matrix pow(ll k) const {
52         assert(n == m);
53         Matrix res(n), base = *this;
54         res.identity();
55         while (k) {
```

```

56         if (k & 1) res = res * base;
57         base = base * base;
58         k >>= 1;
59     }
60     return res;
61 }
62 };

```

K-th permutation

```

1  ll fac[21];
2  void preFac() {
3      fac[0] = 1;
4      for(int i = 1; i <= 20; ++i)
5          fac[i] = fac[i - 1] * i;
6  }
7
8  vector<int> kth_permutation(int n, ll k) { // n^2
9      vector<int> a(n), ans;
10     iota(a.begin(), a.end(), 1);
11     for (int i = n; i >= 1; i--) {
12         ll f = fac[i - 1];
13         ans.push_back(a[k / f]);
14         a.erase(a.begin() + k / f);
15         k %= f;
16     }
17     return ans;
18 }

```

Permutation Index

```

1  ll permutation_index(vector<int>& p) { // n^2
2      int n = int(p.size());
3      vector<int> a(n);
4      iota(a.begin(), a.end(), 1);
5
6      ll k = 0;
7      for (int i = 0; i < n; ++i) {
8          int j = int(find(a.begin(), a.end(), p[i]) - a.begin());
9          k += j * fac[n - 1 - i];
10         a.erase(a.begin() + j);
11     }
12     return k;
13 }

```

Berlekamp Massey

```

1  const ll MOD = 1e9+7;
2  ll add(ll a, ll b) { return (a + b) % MOD; }
3  ll sub(ll a, ll b) { return ((a - b) % MOD + MOD) % MOD; }
4  ll mul(ll a, ll b) { return (a * b) % MOD; }
5  ll power(ll base, ll exp) {
6      ll res = 1;
7      base %= MOD;
8      while (exp > 0) {

```

```

9         if (exp % 2 == 1) res = mul(res, base);
10        base = mul(base, base);
11        exp /= 2;
12    }
13    return res;
14 }
15 ll modInverse(ll n) {
16     return power(n, MOD - 2); // Valid only if MOD is prime
17 }
18 // Finds the shortest linear recurrence transition array for a given sequence S
19 vector<ll> BerlekampMassey(const vector<ll>& S) {
20     int n = S.size(), L = 0, m = 0;
21     vector<ll> C(n), B(n), T;
22     C[0] = B[0] = 1;
23     ll b = 1;
24
25     for (int i = 0; i < n; i++) {
26         ++m;
27         ll d = S[i] % MOD;
28         for (int j = 1; j <= L; j++) d = add(d, mul(C[j], S[i - j]));
29         if (d == 0) continue;
30
31         T = C;
32         ll coef = mul(d, modInverse(b));
33         for (int j = m; j < n; j++) C[j] = sub(C[j], mul(coef, B[j - m]));
34
35         if (2 * L > i) continue;
36         L = i + 1 - L;
37         B = T;
38         b = d;
39         m = 0;
40     }
41     C.resize(L + 1);
42     C.erase(C.begin());
43     for (auto &x : C) x = sub(0, x);
44     return C;
45 }
46 // Multiplies two polynomials modulo the characteristic polynomial (tr)
47 vector<ll> combine(int n, const vector<ll>& a, const vector<ll>& b, const vector<ll>& tr) {
48     vector<ll> res(n * 2 + 1, 0);
49     for (int i = 0; i < n + 1; i++) {
50         for (int j = 0; j < n + 1; j++) {
51             res[i + j] = add(res[i + j], mul(a[i], b[j]));
52         }
53     }
54     for (int i = 2 * n; i > n; --i) {
55         for (int j = 0; j < n; j++) {
56             res[i - 1 - j] = add(res[i - 1 - j], mul(res[i], tr[j]));
57         }
58     }
59     res.resize(n + 1);
60     return res;
61 }
62 // S contains the initial sequence values, 'tr' is the transition array from BM.
63 // K is 0-indexed, S and tr must be same size
64 ll LinearRecurrence(const vector<ll>& S, const vector<ll>& tr, ll k) { //  $O(N^2 \log K)$ 

```

```

65     assert(S.size() == tr.size());
66     int n = tr.size();
67     if (n == 0) return 0;
68     if (k < n) return S[k] % MOD;
69
70     vector<ll> pol(n + 1), e(pol);
71     pol[0] = e[1] = 1;
72
73     for (++k; k; k /= 2) {
74         if (k % 2) pol = combine(n, pol, e, tr);
75         e = combine(n, e, e, tr);
76     }
77
78     ll res = 0;
79     for (int i = 0; i < n; i++)
80         res = add(res, mul(pol[i + 1], S[i]));
81     return res;
82 }

```


5. Strings

Trie (s)

Global Arrays

```
1  const int MXN = 1000005, S = 26, OFFSET = 'a';
2  int nxt[MXN][S], cnt[MXN], isend[MXN], nodes = 1;
3  // clear the arrays you add here in clear() function
4  void clear() {
5      for (int i = 0; i < nodes; i++) {
6          cnt[i] = isend[i] = 0;
7          memset(nxt[i], 0, sizeof(nxt[i]));
8      }
9      nodes = 1;
10 }
11 void insert(const string& s) {
12     int u = 0;
13     for (char c : s) {
14         int v = c - OFFSET;
15         if (!nxt[u][v]) nxt[u][v] = nodes++;
16         u = nxt[u][v];
17         cnt[u]++;
18     }
19     isend[u]++;
20 }
21 int search(const string& s) {
22     int u = 0;
23     for (char c : s) {
24         int v = c - OFFSET;
25         if (!nxt[u][v]) return false;
26         u = nxt[u][v];
27     }
28     return cnt[u];
29 }
30 // clear() insert()
```

Unordered_map

```
1  const int MXN = 1000005;
2  struct Node {
3      unordered_map<char, int> nxt;
4      int cnt = 0;
5  } tree[MXN];
6  int node_cnt = 1;
7  void clear() {
8      for (int i = 0; i < node_cnt; i++) {
9          tree[i].nxt.clear();
10         tree[i].cnt = 0;
11     }
12     node_cnt = 1;
13 }
14 void insert(const string& s) {
15     int u = 0;
16     for (char c : s) {
17         auto it = tree[u].nxt.find(c);
18         if (it == tree[u].nxt.end()) {
```

```

19         tree[u].nxt[c] = node_cnt;
20         u = node_cnt++;
21     } else {
22         u = it->second;
23     }
24     tree[u].cnt++;
25 }
26 }
27 int search(const string& s) {
28     int u = 0;
29     for (char c : s) {
30         auto it = tree[u].nxt.find(c);
31         if (it == tree[u].nxt.end()) return false;
32         u = it->second;
33     }
34     return tree[u].cnt;
35 }
36 // clear() insert()

```

Forward Start Trie

```

1  const int MXN = 1000005;
2  int head[MXN]; // points to the first child of a node
3  int nxt[MXN]; // points to the next sibling of a node
4  char val[MXN]; // the character that leads to this node
5  int cnt[MXN], isend[MXN];
6  int node_cnt = 1;
7  void clear() {
8      fill(head, head + node_cnt, 0);
9      fill(nxt, nxt + node_cnt, 0);
10     fill(cnt, cnt + node_cnt, 0);
11     fill(isend, isend + node_cnt, 0);
12     node_cnt = 1;
13 }
14 void insert(const string& s) {
15     int u = 0;
16     for (char c : s) {
17         int child = head[u];
18         bool found = false;
19         while (child) {
20             if (val[child] == c) {
21                 u = child;
22                 found = true;
23                 break;
24             }
25             child = nxt[child];
26         }
27         if (!found) {
28             val[node_cnt] = c;
29             nxt[node_cnt] = head[u];
30             head[u] = node_cnt;
31             u = node_cnt++;
32         }
33         cnt[u]++;
34     }
35 }
36 int search(const string& s) {

```

```

37     int u = 0;
38     for (char c : s) {
39         int child = head[u];
40         bool found = false;
41         while (child) {
42             if (val[child] == c) {
43                 u = child;
44                 found = true;
45                 break;
46             }
47             child = nxt[child];
48         }
49         if (!found) return 0;
50     }
51     return cnt[u];
52 }

```

Rolling Hash

```

1  using u64 = uint64_t;
2  mt19937_64 rng(chrono::steady_clock::now().time_since_epoch().count() ^
3      (uintptr_t)make_unique<char>().get());
4
5  struct hash61 {
6      static const u64 md = (1LL << 61) - 1;
7      inline static u64 step = (md >> 2) + rng() % (md >> 1);
8      inline static vector<u64> pw = {1};
9
10     int n;
11     vector<u64> pref, suff;
12
13     u64 add(u64 a, u64 b) const { return (a += b) >= md ? a - md : a; }
14     u64 sub(u64 a, u64 b) const { return (a += md - b) >= md ? a - md : a; }
15     u64 mul(u64 a, u64 b) const { return __uint128_t(a) * b % md; }
16
17     template<class T>
18     hash61(const T& s) : n(s.size()), pref(n + 1), suff(n + 1) {
19         while (pw.size() <= n) pw.push_back(mul(pw.back(), step));
20         pref[0] = suff[n] = 1;
21         for (int i = 0; i < n; i++) pref[i + 1] =
22             add(mul(pref[i], step), s[i]);
23         for (int i = n - 1; i >= 0; i--) suff[i] =
24             add(mul(suff[i + 1], step), s[i]);
25     }
26
27     u64 operator()(int l, int r) const { return sub(pref[r + 1],
28         mul(pref[l], pw[r - l + 1])); }
29     u64 rev(int l, int r) const { return sub(suff[l],
30         mul(suff[r + 1], pw[r - l + 1])); }
31 };
32
33 struct hash61 {
34     static const u64 md = (1LL << 61) - 1;
35     inline static u64 step = (md >> 2) + rng() % (md >> 1);
36
37     inline static u64 power(u64 b, u64 e) {

```

```

38     u64 r = 1;
39     while (e) {
40         if (e & 1) r = (unsigned __int128)r * b % md;
41         b = (unsigned __int128)b * b % md;
42         e >>= 1;
43     }
44     return r;
45 }
46
47 inline static u64 inv_step = power(step, md - 2);
48 inline static vector<u64> pw = {1}, ipw = {1};
49
50 int n = 0;
51 vector<u64> pref = {1}, suff = {0};
52
53 u64 add(u64 a, u64 b) const { return (a += b) >= md ? a - md : a; }
54 u64 sub(u64 a, u64 b) const { return (a += md - b) >= md ? a - md : a; }
55 u64 mul(u64 a, u64 b) const { return (unsigned __int128)a * b % md; }
56
57 hash61() {}
58 template<class T> hash61(const T& s) { for (auto c : s) push_back(c); }
59
60 void push_back(u64 c) {
61     while (pw.size() <= n + 1) {
62         pw.push_back(mul(pw.back(), step));
63         ipw.push_back(mul(ipw.back(), inv_step));
64     }
65     pref.push_back(add(mul(pref.back(), step), c));
66     suff.push_back(add(suff.back(), mul(c, pw[n])));
67     n++;
68 }
69
70 void pop_back() {
71     if (!n) return;
72     pref.pop_back();
73     suff.pop_back();
74     n--;
75 }
76
77 u64 operator()(int l, int r) const { return sub(pref[r + 1], mul(pref[l],
78     pw[r - l + 1])); }
79 u64 rev(int l, int r) const { return mul(sub(suff[r + 1], suff[l]), ipw[l]); }
80 };

```

KMP

```

1  vector<int> KMP(const string &s){
2      int n = s.length();
3      vector<int> pi(n);
4      for (int i = 1; i < n; i++){
5          int j = pi[i - 1];
6          while (j > 0 && s[i] != s[j]){
7              j = pi[j - 1];
8          }
9          if (s[i] == s[j]) j++;
10         pi[i] = j;

```

```

11     }
12     return pi;
13 }
14 vector<int> find_occurrences(const string &pat, const string &text){
15     string s = pat + '#' + text;
16     vector<int> pi = KMP(s), ret;
17     int m = pat.size();
18     for (int i = m + 1; i < (int)s.size(); i++){
19         if (pi[i] == m) ret.push_back(i - 2 * m);
20     }
21     return ret;
22 }
23 vector<vector<int>> automaton(string s){ // aut[s.length() + 1][26]
24     s += '#';
25     int n = s.length();
26     vector<int> pi = KMP(s);
27     vector<vector<int>> aut(n, vector<int>(26));
28     for (int i = 0; i < n; i++){
29         for (int j = 0; j < 26; j++){
30             if (i > 0 && 'a' + j != s[i]){
31                 aut[i][j] = aut[pi[i - 1]][j];
32             }else{
33                 aut[i][j] = i + ('a' + j == s[i]);
34             }
35         }
36     }
37     return aut;
38 }

```

Z-Algo

```

1 vector<int> z_algo(string s) {
2     vector<int> z(s.size());
3     for(int i = 1, l = 0, r = 0; i < s.size(); i++) {
4         if(i < r) z[i] = min(r - i, z[i - l]);
5         while(i + z[i] < s.size() && s[z[i]] == s[z[i] + i]) z[i]++;
6         if(i + z[i] > r) r = i + z[i], l = i;
7     }
8     return z;
9 }

```

Largest Lexicographical Substring

```

1 string largestLexSubstring(const string &s) {
2     int n = int(s.size());
3     int i = 0, j = 1, k = 0;
4
5     while (j + k < n) {
6         if (s[i + k] == s[j + k]) k++;
7         else if (s[i + k] < s[j + k])
8             /* change it to > if you want lowest */
9             i = max(i + k + 1, j), j = i + 1, k = 0;
10        else j = j + k + 1, k = 0;
11    }
12

```

```

13     return s.substr(i);
14 }

```

Manacher

```

1  auto manacher(const string &t) {
2      string s = "%#";
3      s.reserve(t.size() * 2 + 3);
4      for(char c : t) s += c + "#";
5      s += '$';
6      // t = aabaacaabaa -> s = %a#a#b#a#a#c#a#a#b#a#a#$
7
8      vector<int> res(s.size());
9      for(int i = 1, l = 1, r = 1; i < s.size(); i++) {
10         res[i] = max(0, min(r - i, res[l + r - i]));
11         while(s[i + res[i]] == s[i - res[i]]) res[i]++;
12         if(i + res[i] > r) {
13             l = i - res[i];
14             r = i + res[i];
15         }
16     }
17     for(auto &i : res) i--;
18     return vector(res.begin() + 2, res.end() - 2); // a#a#b#a#a#c#a#a#b#a#a
19     // get max odd len = res[2 * i]; aba -> i = b
20     // get max even len = res[2 * i + 1]; abba -> i = first b
21 }

```

Aho Corasick Algorithm

```

1  namespace corasick {
2      const int N = +++++;
3      const int SIGMA = +++++;
4
5      int nxt[N][SIGMA], fail_link[N], dict_link[N], match_idx[N], nodes;
6
7      void init() {
8          nodes = fail_link[0] = dict_link[0] = 0;
9          memset(nxt[0], 0, sizeof(nxt[0]));
10         match_idx[0] = -1;
11         nodes++;
12     }
13
14     int create_node() {
15         memset(nxt[nodes], 0, sizeof(nxt[nodes]));
16         fail_link[nodes] = dict_link[nodes] = 0;
17         match_idx[nodes] = -1;
18         return nodes++;
19     }
20
21     int insert(const string& pattern, int id) {
22         int cur = 0;
23         for (char c : pattern) {
24             if (!nxt[cur][c]) {
25                 nxt[cur][c] = create_node();
26             }
27             cur = nxt[cur][c];

```

```

28     }
29     if (~match_idx[cur]) return match_idx[cur];
30     return match_idx[cur] = id;
31 }
32
33 void build() {
34     queue<int> q;
35
36     for (int c = 0; c < SIGMA; ++c) {
37         int chi = nxt[0][c];
38         if (chi) {
39             fail_link[chi] = dict_link[chi] = 0;
40             q.push(chi);
41         }
42     }
43
44     while (!q.empty()) {
45         int u = q.front();
46         q.pop();
47
48         for (int c = 0; c < SIGMA; ++c) {
49             int chi = nxt[u][c];
50
51             if (chi) {
52                 fail_link[chi] = nxt[fail_link[u]][c];
53
54                 int fail = fail_link[chi];
55                 if (match_idx[fail] != -1)
56                     dict_link[chi] = fail;
57                 else
58                     dict_link[chi] = dict_link[fail];
59                 q.push(chi);
60             } else
61                 nxt[u][c] = nxt[fail_link[u]][c];
62         }
63     }
64 }
65
66 vector<vector<int>> search(const string& text, const vector<int>& lengths) {
67     vector<vector<int>> ret(lengths.size());
68     for (int cur = 0, i = 0; i < text.length(); ++i) {
69         cur = nxt[cur][text[i]];
70
71         for (int u = cur; u; u = dict_link[u]) {
72             int id = match_idx[u];
73             if (id != -1) {
74                 ret[id].push_back(i - lengths[id] + 1);
75             }
76         }
77     }
78     return ret;
79 }
80 } /* init() insert() build() search() */

```

Dynamic Aho Corasick

```

1 struct AC {

```

```

2   int N, P;
3   static const int A = 26;
4   vector<array<int, A> > next;
5   vector<int> link, out_link, cnt;
6   vector<vector<int> > out, rev;
7   AC() : N(0), P(0) { node(); }
8
9   int node() {
10      next.push_back(array<int, A>{0});
11      rev.push_back({});
12      cnt.emplace_back(0);
13      link.emplace_back(0);
14      out_link.emplace_back(0);
15      out.emplace_back();
16      return N++;
17  }
18
19  inline int get(char c) { return c - 'a'; }
20
21  int add_pattern(const string &T) {
22      int u = 0;
23      for (auto c: T) {
24          if (!next[u][get(c)]) next[u][get(c)] = node();
25          u = next[u][get(c)];
26      }
27      out[u].push_back(P);
28      cnt[u]++;
29      return P++;
30  }
31
32  void build() {
33      queue<int> q;
34      q.push(0);
35      while (!q.empty()) {
36          int u = q.front();
37          q.pop();
38          for (int c = 0; c < A; c++) {
39              int v = next[u][c];
40              if (!v) next[u][c] = next[link[u]][c];
41              else {
42                  link[v] = u ? next[link[u]][c] : 0;
43                  rev[link[v]].push_back(v);
44                  cnt[v] += cnt[link[v]];
45                  out_link[v] = out[link[v]].empty() ?
46                      out_link[link[v]] : link[v];
47                  q.push(v);
48              }
49          }
50      }
51  }
52
53  int advance(int u, char c) {
54      return next[u][get(c)];
55  }
56
57  vector<vector<int> > search(const string &s, vector<int> &pat) {

```



```

58     int n = (int) s.length(), m = (int) pat.size();
59     int u = 0;
60     vector<vector<int> > ret(m);
61     for (int i = 0; i < n; i++) {
62         char c = s[i];
63         u = advance(u, c);
64         int x = u;
65         while (x != 0) {
66             for (int v: out[x]) ret[v].push_back(i - pat[v] + 1);
67             x = out_link[x];
68         }
69     }
70     return ret;
71 }
72 };
73
74 struct dynamic_AC {
75     int lg;
76     vector<vector<string> > a;
77     vector<AC> aho;
78
79     dynamic_AC(int LOG = 20) {
80         lg = LOG;
81         a = vector<vector<string> >(lg);
82         aho = vector<AC>(lg);
83     }
84
85     void insert(const string &s) {
86         vector<string> have = {s};
87         for (int i = 0; i < lg; i++) {
88             if (i == lg - 1) {
89                 lg++;
90                 a.push_back({});
91                 aho.push_back(AC());
92             }
93             if (a[i].empty()) {
94                 swap(have, a[i]);
95                 for (auto &v: a[i]) aho[i].add_pattern(v);
96                 aho[i].build();
97                 break;
98             } else {
99                 have.insert(end(have), begin(a[i]), end(a[i]));
100                 vector<string>().swap(a[i]);
101                 aho[i] = AC();
102             }
103         }
104     }
105
106     ll cnt(const string &s) {
107         ll answer = 0;
108         for (int i = 0; i < lg; i++) {
109             int state = 0;
110             for (auto &v: s) {
111                 state = aho[i].advance(state, v);
112                 answer += aho[i].cnt[state];
113             }
114         }

```

```

115         return answer;
116     }
117 };

```

Suffix Array

```

1 // O(n log(n))
2 struct suffix {
3     int n;
4     vector<int> p, c, lcp;
5     string s;
6
7     explicit suffix(string _s) : n(int(_s.size()) + 1), s(move(_s)), p(n),
8         c(n), lcp(n - 1) {
9         s += char(0);
10        iota(p.begin(), p.end(), 0);
11        sort(p.begin(), p.end(), [&](int i, int j) { return s[i] < s[j]; });
12        for (int i = 1; i < n; i++) c[p[i]] = c[p[i - 1]] + (s[p[i]] != s[p[i - 1]]);
13        vector<int> nc(n), np(n);
14        int k = 1;
15        while (k < n) {
16            vector<int> f(n + 1);
17            for (int i = 0; i < n; i++) p[i] = (p[i] - k + n) % n, f[c[i] + 1]++;
18            for (int i = 1; i <= n; i++) f[i] += f[i - 1];
19            for (int i = 0; i < n; i++) np[f[c[p[i]]]++] = p[i];
20            swap(p, np), nc[p[0]] = 0;
21            for (int i = 1; i < n; i++)
22                nc[p[i]] = nc[p[i - 1]] +
23                    (c[p[i]] != c[p[i - 1]]
24                     || c[(p[i] + k) % n] != c[(p[i - 1] + k) % n]);
25            swap(c, nc), k <= 1;
26        }
27        for (int i = k = 0; i < n - 1; i++) {
28            int j = p[c[i] - 1];
29            for (; s[i + k] == s[j + k]; k++);
30            if (c[i]) lcp[c[i] - 1] = k;
31            k = max(k - 1, 0);
32        }
33        s.pop_back(), c.pop_back(), n--;
34        p.erase(p.begin()), lcp.erase(lcp.begin());
35        for (auto &i : c) i--;
36    }
37    vector<vector<int>> table;
38    void buildLcp() {
39        int LOG = __lg(n) + 1;
40        table.resize(LOG, vector<int>(n));
41        table[0] = lcp;
42        for (int l = 1; l < LOG; l++) {
43            for (int i = 0; i + (1 << (l - 1)) < n; i++) {
44                table[l][i] = min(table[l - 1][i], table[l - 1][i + (1 << (l - 1))]);
45            }
46        }
47    }
48    int query(int l, int r) { // 0-based
49        if (l == r) return n - 1 - l;
50        l = c[l], r = c[r];

```

```

51         if(l > r) swap(l, r);
52         r--;
53         int len = __lg(r - l + 1);
54         return min(table[len][l], table[len][r - (1 << len) + 1]);
55     }
56     auto count(string const &t) { // O(log(n) * size(t))
57         int l = int(lower_bound(p.begin(), p.end(), -1, [&](int i, int j) {
58             return s.substr(i, min<size_t>(t.size(), n - i)) < t;
59         }) - p.begin());
60         int r = int(lower_bound(p.begin() + l, p.end(), -1, [&](int i, int j) {
61             return s.substr(i, min<size_t>(t.size(), n - i)) <= t;
62         }) - p.begin()) - 1;
63         return pair{l, r};
64     }
65 };

```

Suffix Automaton

```

1  namespace sam {
2      const int N = 4000200, SIGMA = 26; // N two times the max length
3      int sa[N][SIGMA], len[N], fail[N], terminal[N], nodes, lst, C[N];
4
5      void init() {
6          nodes = lst = 0;
7          memset(sa[0], -1, sizeof sa[0]);
8          fail[0] = -1;
9          len[0] = terminal[0] = 0;
10     }
11
12     int create_node() {
13         nodes++;
14         memset(sa[nodes], -1, sizeof sa[nodes]);
15         terminal[nodes] = 0;
16         return nodes;
17     }
18
19     void insert(char ch) {
20         int u = create_node();
21         len[u] = len[lst] + 1;
22         int p, q;
23         for (p = lst; ~p; p = fail[p]) {
24             q = sa[p][ch];
25             if (~sa[p][ch]) break;
26             sa[p][ch] = u;
27         }
28
29         if (p == -1) {
30             fail[u] = 0;
31         } else if (len[p] + 1 == len[q]) {
32             fail[u] = q;
33         } else {
34             int x = create_node();
35             len[x] = len[p] + 1;
36             memcpy(sa[x], sa[q], sizeof sa[0]);
37             for (int v = p; ~v && sa[v][ch] == q; v = fail[v])
38                 sa[v][ch] = x;

```

```

39         fail[x] = fail[q];
40         fail[q] = fail[u] = x;
41     }
42     lst = u;
43     C[u] = 1;
44 }
45
46 void insert(const string &s) {
47     for (auto ch: s)
48         insert(ch);
49 }
50
51 string walk(const string &s) {
52     int cur = 0;
53     for (auto ch: s) {
54         if (sa[cur][ch] == -1) return "NO"s;
55         cur = sa[cur][ch];
56     }
57     return "YES";
58 }
59
60 void pre_count() {
61     vector<vector<int> > vlen(N);
62     for (int i = 0; i < N; i++) vlen[len[i]].push_back(i);
63     for (int l = N - 1; ~l; l--)
64         for (auto x: vlen[l])
65             C[fail[x]] += C[x];
66 }
67
68 int count(const string &s) {
69     int cur = 0;
70     for (auto ch: s) {
71         if (!~sa[cur][ch]) return 0;
72         cur = sa[cur][ch];
73     }
74     return C[cur];
75 }
76
77 vector<ll> count_distinct() {
78     vector<ll> dp(nodes + 1), outdeg(nodes + 1);
79     vector<vector<int> > indeg(nodes + 1);
80     for (int i = 0; i <= nodes; i++)
81         for (int j = 0; j < SIGMA; j++)
82             if (~sa[i][j]) {
83                 outdeg[i]++;
84                 indeg[sa[i][j]].push_back(i);
85             }
86     queue<int> q;
87     for (int i = 0; i <= nodes; i++)
88         if (!outdeg[i])
89             q.push(i);
90     while (!q.empty()) {
91         int top = q.front();
92         q.pop();
93         dp[top] = 1;
94         for (int i = 0; i < SIGMA; i++)

```

```

95         if (~sa[top][i])
96             dp[top] += dp[sa[top][i]];
97     for (auto x: indeg[top]) {
98         outdeg[x]--;
99         if (!outdeg[x]) q.push(x);
100     }
101 }
102
103     return dp;
104 }
105
106 ll count_distinct2() {
107     ll ret = 0;
108     for (int i = 1; i <= nodes; i++)
109         ret += len[i] - len[fail[i]];
110     return ret;
111 }
112
113 ll sum_distinct() {
114     vector<ll> d(nodes + 1), ans(nodes + 1), outdeg(nodes + 1);
115     vector<vector<int>> indeg(nodes + 1);
116     queue<int> q;
117     for (int i = 0; i <= nodes; i++) {
118         for (int j = 0; j < SIGMA; j++)
119             if (~sa[i][j]) {
120                 outdeg[i]++;
121                 indeg[sa[i][j]].push_back(i);
122             }
123         if (!outdeg[i]) q.push(i);
124     }
125     while (!q.empty()) {
126         int top = q.front();
127         q.pop();
128
129         d[top] = 1;
130         for (int i = 0; i < SIGMA; i++)
131             if (~sa[top][i]) {
132                 d[top] += d[sa[top][i]];
133                 ans[top] += d[sa[top][i]] + ans[sa[top][i]];
134             }
135
136         for (int x: indeg[top])
137             if (!--outdeg[x])
138                 q.push(x);
139     }
140     return ans.front();
141 }
142
143 ll sum_distinct2() {
144     ll ret = 0;
145     for (int i = 1; i <= nodes; i++) {
146         int a = len[i], b = len[fail[i]] + 1;
147         ll num = a-b+1;
148         ret += num * (a+b)/2;
149     }
150     return ret;
151 }

```

```

152
153 string get_kth(ll k) {
154     auto d = count_distinct();
155     string ret;
156     int cur = 0;
157     while (k) {
158         for (int i = 0; i < SIGMA; i++) if (~sa[cur][i]) {
159             if (d[sa[cur][i]] >= k) {
160                 k--;
161                 ret.push_back(i);
162                 cur = sa[cur][i];
163                 break;
164             }
165             k -= d[sa[cur][i]];
166         }
167     }
168     return ret;
169 }
170
171 string smallest_shift(int n) {
172     int cur = 0;
173     string ret;
174     while (n--) {
175         for (int i = 0; i < SIGMA; i++) if (~sa[cur][i]) {
176             cur = sa[cur][i];
177             ret.push_back(i);
178             break;
179         }
180     }
181     return ret;
182 }
183
184 vector<int> vis(N);
185 void dfs(int cur, const vector<ll>& d) {
186     if (vis[cur]) return;
187     for (int i = 0; i < SIGMA; i++) if (~sa[cur][i]) {
188         cout << cur << "," << d[cur]-1 << '□' << sa[cur][i] <<
189             "," << d[sa[cur][i]]-1 << '□' << char(i+'a') << '\n';
190         dfs(sa[cur][i], d);
191     }
192 }
193 void prnt() {
194     auto d = count_distinct();
195     dfs(0, d);
196 }
197 } /* init() insert() */

```

Suffix Automaton Subproblems

6. Dynamic Programming

LIS

```
1  int lis_size(const vector<int>& nums) {
2      vector<int> tail;
3      for (auto x : nums) {
4          auto it = lower_bound(tail.begin(), tail.end(), x);
5          if (it == tail.end()) tail.push_back(x);
6          else *it = x;
7      }
8      return tail.size();
9  }
10
11 int lis_with_sequence(const vector<int>& a,
12                      vector<int>& out_seq)
13 {
14     int n = a.size();
15     vector<int> tail;
16     vector<int> tail_idx;
17     vector<int> prev(n, -1);
18     tail.reserve(n);
19     tail_idx.reserve(n);
20
21     for (int i = 0; i < n; ++i) {
22         int x = a[i];
23         auto it = lower_bound(tail.begin(), tail.end(), x);
24         int pos = it - tail.begin();
25         if (it == tail.end()) {
26             tail.push_back(x);
27             tail_idx.push_back(i);
28         }
29         else {
30             *it = x;
31             tail_idx[pos] = i;
32         }
33         if (pos > 0)
34             prev[i] = tail_idx[pos-1];
35     }
36
37     int lis_len = tail.size();
38     out_seq.clear();
39     for (int cur = tail_idx[lis_len - 1]; cur >= 0; cur = prev[cur])
40         out_seq.push_back(a[cur]);
41     reverse(out_seq.begin(), out_seq.end());
42
43     return lis_len;
44 }
```

0/1 Knapsack

```
1  int knapsack(int W, vector<int> &val, vector<int> &wt) {
2      vector<int> dp(W + 1, 0);
3      for (int i = 1; i <= wt.size(); i++)
4          for (int j = W; j >= wt[i - 1]; j--)
5              dp[j] = max(dp[j], dp[j - wt[i - 1]] + val[i - 1]);
```

```

6
7
8     return dp[W];
9 }
10
11 int unbounded_knapSack(int capacity, vector<int> &val, vector<int> &wt) {
12     vector<int> dp(capacity + 1, 0);
13     for (int i = val.size() - 1; i >= 0; i--) {
14         for (int j = 1; j <= capacity; j++) {
15             int take = 0;
16             if (j - wt[i] >= 0) {
17                 take = val[i] + dp[j - wt[i]];
18             }
19             int noTake = dp[j];
20             dp[j] = max(take, noTake);
21         }
22     }
23     return dp[capacity];
24 }

```

Edit Distance

```

1 string a, b; cin >> a >> b;
2 int na = a.size(), nb = b.size();
3 int dp[na+1][nb+1];
4 memset(dp, 0, sizeof dp);
5
6 for (int i = 1; i <= na; i++)
7     dp[i][0] = i;
8 for (int j = 1; j <= nb; j++)
9     dp[0][j] = j;
10 for (int i = 1; i <= na; i++) {
11     for (int j = 1; j <= nb; j++) {
12         dp[i][j] = min({dp[i-1][j], dp[i][j-1], dp[i-1][j-1]}) + 1;
13         if (a[i-1] == b[j-1])
14             dp[i][j] = min(dp[i][j], dp[i-1][j-1]);
15     }
16 }
17 cout << dp[na][nb];

```

Longest Common Subsequence

```

1 void lcs(char* X, char* Y, int m, int n)
2 {
3     int L[m + 1][n + 1];
4     for (int i = 0; i <= m; i++) {
5         for (int j = 0; j <= n; j++) {
6             if (i == 0 || j == 0)
7                 L[i][j] = 0;
8             else if (X[i - 1] == Y[j - 1])
9                 L[i][j] = L[i - 1][j - 1] + 1;
10            else
11                L[i][j] = max(L[i - 1][j], L[i][j - 1]);
12        }
13    }
14    int index = L[m][n];

```



```

15
16 char lcs[index + 1];
17 lcs[index] = '\0';
18 int i = m, j = n;
19 while (i > 0 && j > 0) {
20     if (X[i - 1] == Y[j - 1]) {
21         lcs[index - 1]
22         = X[i - 1];
23         i--;
24         j--;
25         index--;
26     }
27     else if (L[i - 1][j] > L[i][j - 1])
28         i--;
29     else
30         j--;
31 }
32 }

```

Shortest Common Supersequence

```

1 while (i > 0 && j > 0) {
2     if (str1[i - 1] == str2[j - 1]) {
3         result.push_back(str1[i - 1]);
4         i--;
5         j--;
6     } else if (dp[i - 1][j] > dp[i][j - 1]) {
7         result.push_back(str1[i - 1]);
8         i--;
9     } else {
10        result.push_back(str2[j - 1]);
11        j--;
12    }
13 }
14 while (i > 0) {
15     result.push_back(str1[i - 1]);
16     i--;
17 }
18 while (j > 0) {
19     result.push_back(str2[j - 1]);
20     j--;
21 }
22 reverse(result.begin(), result.end());
23 return result;

```

Count Subsets Sum to K

```

1 int perfectSum(vector<int> &arr, int target) {
2     int n = arr.size();
3     vector<int> dp(target + 1, 0), ndp(target + 1, 0);
4     dp[0] = 1;
5     for (int i = 1; i <= n; i++) {
6         ndp = dp;
7         for (int j = 0; j <= target; j++) {
8             if (j >= arr[i - 1]) {
9                 ndp[j] += dp[j - arr[i - 1]];

```

```

10         }
11     }
12     dp = ndp;
13 }
14 return ndp[target];
15 }

```

Hamiltonian Paths

```

1 int rec(int u, int vis) {
2     vis |= (1 << u);
3     if (u == n-1)
4         return __builtin_popcount(vis) == n;
5
6     int& ans = dp[vis][u];
7     if (~ans) return ans;
8     // dpid[u][vis] = 2;
9     ans = 0;
10    each(ver, g[u])
11        if (!((vis >> ver) & 1))
12            ans = (ans + rec(ver, vis | (1 << ver)))%mod;
13    return ans;
14 }

```

Kadane

```

1 template<class T>
2 array<ll, 3> kadane(const vector<T>& arr) {
3     ll mx = LLONG_MIN, csum = 0;
4     ll s = 0, e = 0, ts = 0, sz = arr.size();
5     for (ll i = 0; i < sz; i++) {
6         if (csum + arr[i] > arr[i]) {
7             csum += arr[i];
8         } else {
9             csum = arr[i];
10            ts = i;
11        }
12        if (csum > mx) {
13            mx = csum;
14            s = ts;
15            e = i;
16        }
17    }
18    return {mx, s, e};
19 }

```

Dynamic max subarray sum

```

1 struct info {
2     int sum, pref, suff, ans, mnelement = -1e4-10;;
3     info(int x) {
4         sum = pref = suff = x;
5         // ans = max<int>(x, 0);
6         ans = x; // if empty subarray is not allowed
7     }

```

```

8     info() { // default value
9         sum = pref = suff = ans = mnelement;
10    }
11    friend info operator+(const info &l, const info &r) {
12        info ret;
13        ret.sum = l.sum + r.sum;
14        ret.pref = max(l.pref, l.sum + r.pref);
15        ret.suff = max(r.suff, r.sum + l.suff);
16        ret.ans = max({l.ans, r.ans, l.suff + r.pref});
17        return ret;
18    }
19 };

```

Meet in the middle

```

1  vector<int> a;
2  auto get_subset_sums = [&](int l, int r) -> vector<ll> {
3      int len = r - l + 1;
4      vector<ll> res;
5      for (int i = 0; i < (1 << len); i++) {
6          ll sum = 0;
7          for (int j = 0; j < len; j++)
8              if (i & (1 << j))
9                  sum += a[l + j];
10
11         res.push_back(sum);
12     }
13     return res;
14 };
15 vector<ll> left = get_subset_sums(0, n / 2 - 1);
16 vector<ll> right = get_subset_sums(n / 2, n - 1);
17 sort(left.begin(), left.end());
18 sort(right.begin(), right.end());
19 ll ans = 0;
20 for (ll i : left) {
21     auto low_iterator = lower_bound(right.begin(), right.end(), x - i);
22     auto high_iterator = upper_bound(right.begin(), right.end(), x - i);
23     ans += high_iterator - low_iterator;
24 }

```

Digit DP

```

1  using ull = unsigned long long int;
2  #define int ull
3  pair<int, int> dp[2][16];
4  bool vis[2][16];
5  string num;
6  pair<int, int> rec(int idx, bool U) {
7      if (idx == num.size()) return {0, 1};
8      pair<int, int> &ret = dp[U][idx];
9      if (vis[U][idx]) return ret;
10     vis[U][idx] = 1;
11     ret = {0, 0};
12     int D = U ? num[idx] - '0' : 9;
13     for (int d = 0; d <= D; d++) {
14         auto [x, y] = rec(idx+1, U&d==D);

```

```
15         ret.first += x + d * y;  
16         ret.second += y;  
17     }  
18     return ret;  
19 }
```

7. Bit Manipulation

Binary Trie (s)

```
1 // MXN should generally be (Max Queries * MXB)
2 const int MXN = 500500 * 30, S = 2, MXB = 30;
3 int nxt[MXN][S], cnt[MXN], nodes = 1;
4 // clear the arrays you add here in clear() function
5 void clear() {
6     for (int i = 0; i < nodes; i++) {
7         nxt[i][0] = nxt[i][1] = 0;
8         cnt[i] = 0;
9     }
10    nodes = 1;
11 }
12 void trieinsert(long long val) {
13     int u = 0;
14     for (int i = MXB - 1; i >= 0; i--) {
15         int bit = (val >> i) & 1;
16         if (!nxt[u][bit]) nxt[u][bit] = nodes++;
17         u = nxt[u][bit];
18         cnt[u]++;
19     }
20 }
21 int triecount(long long val) {
22     int u = 0;
23     for (int i = MXB - 1; i >= 0; i--) {
24         int bit = (val >> i) & 1;
25         if (!nxt[u][bit]) return 0;
26         u = nxt[u][bit];
27     }
28     return cnt[u];
29 }
30 void trieerase(long long val) {
31     if (!triecount(val)) return;
32     int u = 0;
33     for (int i = MXB - 1; i >= 0; i--) {
34         int bit = (val >> i) & 1;
35         u = nxt[u][bit];
36         cnt[u]--;
37     }
38 }
39 long long get_max_xor(long long val) {
40     if (nodes == 1 || cnt[nxt[0][0]] + cnt[nxt[0][1]] == 0) return -1;
41     int u = 0;
42     long long ans = 0;
43     for (int i = MXB - 1; i >= 0; i--) {
44         int bit = (val >> i) & 1;
45         int opp = bit ^ 1;
46         if (nxt[u][opp] && cnt[nxt[u][opp]] > 0) {
47             ans |= (1LL << i);
48             u = nxt[u][opp];
49         } else {
50             u = nxt[u][bit];
51         }
52     }
53     return ans;
```

```

54 }
55 long long get_min_xor(long long val) {
56     if (nodes == 1 || cnt[nxt[0][0]] + cnt[nxt[0][1]] == 0) return -1;
57     int u = 0;
58     long long ans = 0;
59     for (int i = MXB - 1; i >= 0; i--) {
60         int bit = (val >> i) & 1;
61         if (nxt[u][bit] && cnt[nxt[u][bit]] > 0) {
62             u = nxt[u][bit];
63         } else {
64             ans |= (1LL << i);
65             u = nxt[u][bit ^ 1];
66         }
67     }
68     return ans;
69 }
70 // Returns the count of numbers 'x' in the trie such that (x ^ val) < k
71 int query(long long val, long long k) {
72     int u = 0, ans = 0;
73     for (int i = MXB - 1; i >= 0; i--) {
74         if (!u) break;
75         int v_bit = (val >> i) & 1;
76         int k_bit = (k >> i) & 1;
77
78         if (k_bit == 1) {
79             // If k's bit is 1, taking v_bit makes the XOR bit 0, which is strictly less than
80             // We add all elements in that subtree to our answer.
81             if (nxt[u][v_bit]) ans += cnt[nxt[u][v_bit]];
82             // Then we traverse down the opposite branch to evaluate lower bits.
83             u = nxt[u][v_bit ^ 1];
84         } else {
85             // If k's bit is 0, we MUST take v_bit to keep the XOR bit 0.
86             u = nxt[u][v_bit];
87         }
88     }
89     return ans;
90 }
91 // clear() insert(), remove() query()

```

Xor Basis

```

1  template<const int Log = 30, typename T = ll>
2  struct basis {
3      int sz = 0;
4      array<T, Log> a{};
5      void add(T x) {
6          if (sz == Log) return;
7          int i;
8          while (x) {
9              if (!a[i = __lg(x)]) return sz++, void(a[i] = x);
10             x ^= a[i];
11         }
12     }
13     T reduce(T x) {
14         if (sz == Log) return 0;
15         T res = 0;

```

```

16     int i;
17     while(x) {
18         if(a[i = __lg(x)]) x ^= a[i];
19         else res |= T(1) << i, x ^= T(1) << i;
20     }
21     return res;
22 }
23 bool find(T x) {
24     if(sz == Log) return true;
25     int i;
26     while(x) {
27         if(a[i = __lg(x)]) x ^= a[i];
28         else return false;
29     }
30     return true;
31 }
32 void clear() {
33     if(sz) a.fill(0), sz = 0;
34 }
35 T getMax(T r = 0) {
36     for(int i = Log - 1; i >= 0; i--) r = max(r ^ a[i], r);
37     return r;
38 }
39
40 T find_k(size_t k, T base_val = 0) {
41     assert(k < 1ULL << sz);
42     T curr = base_val;
43     for(int i = Log - 1, b = sz - 1; i >= 0; i--) {
44         if(a[i]) {
45             if((k >> b & 1) ^ (curr >> i & 1)) curr ^= a[i];
46             b--;
47         }
48     }
49     return curr;
50 }
51
52 T getMaxBounded(T limit, T r = 0) {
53     T best = -1;
54     for (int i = Log - 1; i >= 0; i--) {
55         if (a[i]) {
56             T r1 = r, r2 = r ^ a[i];
57             if ((r1 >> i) & 1) swap(r1, r2);
58
59             if ((limit >> i) & 1) {
60                 T temp = r1;
61                 for (int j = i - 1; j >= 0; j--)
62                     temp = max(temp ^ a[j], temp);
63                 best = max(best, temp);
64                 r = r2;
65             } else
66                 r = r1;
67         } else {
68             int bit_r = (r >> i) & 1;
69             int bit_l = (limit >> i) & 1;
70
71             if (bit_l == 1 && bit_r == 0) {

```

```

72         T temp = r;
73         for (int j = i - 1; j >= 0; j--)
74             temp = max(temp ^ a[j], temp);
75         best = max(best, temp);
76         r = -1;
77         break;
78     }
79     if (bit_l == 0 && bit_r == 1) {
80         r = -1;
81         break;
82     }
83 }
84 }
85 if (r != -1) best = max(best, r);
86
87 return best;
88 }
89
90 friend basis operator+(basis const &l, basis const &r) {
91     if(l.sz == Log) return l;
92     if(r.sz == Log) return r;
93     auto res = l;
94     for(int i = 0; i < Log; i++) if(r.a[i]) res.add(r.a[i]);
95     return res;
96 }
97 };

```

N-SAT

```

1  struct nsat {
2      int n;
3      vector<vector<int>> lit, g;
4      vector<int> occ, pos, neg, val, stk, lvl;
5
6      nsat(int n) : n(n), g(2 * n), occ(2 * n), val(n, -1) {}
7
8      // Pass a list of pairs: {variable_index, is_true}
9      // Example: { {0, 1}, {1, 0} } -> (x_0 OR NOT x_1)
10     void add_clause(initializer_list<pair<int, int>> vars) {
11         vector<int> c;
12         for (auto [u, is_true] : vars) {
13             int L = (u << 1) | (!is_true);
14             c.push_back(L);
15             g[L].push_back(lit.size());
16             occ[L]++;
17         }
18         lit.push_back(c);
19     }
20
21     void apply(int L) {
22         val[L >> 1] = !(L & 1);
23         stk.push_back(L);
24         for (int c : g[L]) {
25             if (pos[c]++ == 0)
26                 for (int u : lit[c]) occ[u]--;
27         }
28     }
29 };

```



```

28     for (int c : g[L ^ 1]) neg[c]++;
29 }
30
31 void undo() {
32     int L = stk.back();
33     stk.pop_back();
34     val[L >> 1] = -1;
35     for (int c : g[L]) {
36         if (--pos[c] == 0)
37             for (int u : lit[c]) occ[u]++;
38     }
39     for (int c : g[L ^ 1]) neg[c]--;
40 }
41
42 bool deduce(int &q_head) {
43     while (q_head < stk.size()) {
44         int L = stk[q_head++];
45         for (int c : g[L ^ 1]) {
46             if (pos[c] > 0) continue; // Clause already satisfied
47             if (neg[c] == lit[c].size()) return false; // Conflict found
48
49             if (neg[c] + 1 == lit[c].size()) { // Unit clause triggered
50                 for (int u : lit[c]) {
51                     if (val[u >> 1] == -1) {
52                         apply(u);
53                         break;
54                     }
55                 }
56             }
57         }
58     }
59     return true;
60 }
61
62 bool ok() {
63     pos.assign(lit.size(), 0);
64     neg.assign(lit.size(), 0);
65     val.assign(n, -1);
66     stk.clear();
67     lvl.clear();
68     int q_head = 0;
69
70     // Force evaluate base unit clauses
71     for (int c = 0; c < lit.size(); ++c) {
72         if (lit[c].empty()) return false;
73         if (lit[c].size() == 1) {
74             int L = lit[c][0];
75             if (val[L >> 1] == -1) apply(L);
76             else if (val[L >> 1] != !(L & 1)) return false;
77         }
78     }
79
80     // Sort variables by frequency for fast O(1) branching
81     vector<int> order(n);
82     iota(order.begin(), order.end(), 0);
83     sort(order.begin(), order.end(), [&](int a, int b) {

```

```

84         return occ[a << 1] + occ[a << 1 | 1] > occ[b << 1] + occ[b << 1 | 1];
85     });
86
87     while (true) {
88         if (deduce(q_head)) {
89             // Find next unassigned variable to branch on
90             int s = -1;
91             for (int v : order) {
92                 if (val[v] == -1) { s = v; break; }
93             }
94
95             // If all variables are assigned, we found a satisfying assignment
96             if (s == -1) return true;
97
98             lvl.push_back(stk.size());
99             apply(s << 1); // Guess that variable 's' is True
100         } else {
101             // Conflict! Backtrack to last decision
102             if (lvl.empty()) return false;
103
104             int backtrack_idx = lvl.back();
105             lvl.pop_back();
106
107             int last_guess = stk[backtrack_idx];
108             while (stk.size() > backtrack_idx) undo();
109
110             q_head = stk.size();
111             apply(last_guess ^ 1); // Flip the guess
112         }
113     }
114 }
115 };

```

Max Xor Subset In Range

```

1 // Given queries L,R find max XOR subset in range
2 const int N = 5e5 + 5;
3 vector<pair<int,int>>qqq[N];
4 int ans[N],basis[22],arr[N],last[22];
5
6 void add(int ind,int x){
7     for (int i = 21; i >= 0; --i) {
8         if((x >> i & 1) == 0)continue;
9         if(ind > last[i]){
10             swap(x,basis[i]);
11             swap(ind,last[i]);
12         }
13         x ^= basis[i];
14     }
15 }
16 int query(int ind){
17     int ret = 0;
18     for (int i = 21; i >= 0; --i) {
19         if(last[i] >= ind){
20             ret = max(ret,ret ^ basis[i]);
21         }

```

```

22     }
23     return ret;
24 }
25
26 void solve() {
27     int n;cin >> n;
28     memset(last,-1,sizeof last);
29     for (int i = 0; i < n; ++i) {
30         cin >> arr[i];
31     }
32     int q;cin >> q;
33     for (int i = 0; i < q; ++i) {
34         int l,r;cin >> l >> r;
35         l--,r--;
36         qq[r].emplace_back(l,i);
37     }
38     for (int i = 0; i < n; ++i) {
39         add(i,arr[i]);
40         for(auto &x:qq[i]){
41             int l = x.fi;
42             int ind = x.se;
43             ans[ind] = query(l);
44         }
45     }
46     for (int i = 0; i < q; ++i) {
47         cout << ans[i] << endl;
48     }
49 }

```

and (&) in range [l, r]

```

1  ll andRange(ll l, ll r) {
2      ll ans=0, msb = -1;
3      while(r>0) { r >>= 1; msb++;}
4      for(ll i= msb; ~i; i--) {
5          ll la = 1;
6          if((l&(la << i)) == (r&(la << i)))
7              ans += (l&(la << i));
8          else
9              break;
10     }
11     return ans;
12 }

```

Bit Twiddle Permute

```

1  int bit_twiddle_permute(int v) { // next integer that has _pop_count(v) bits
2      int t = v | (v - 1);
3      int w = (t + 1) | (((~t & -~t) - 1) >> (__builtin_ctz(v) + 1));
4      return w;
5  }

```

8. Game Theory and Sequences

Mex Calculator

```
1 class mex_calculator {
2     map<int, int> count;
3     set<int> missing;
4     public:
5     mex_calculator(const vector<int>& arr, int upper_bound) { // n log n
6         for (int x : arr)
7             count[x]++;
8         for (int i = 0; i <= upper_bound + 1; ++i)
9             if (count[i] == 0)
10                missing.insert(i);
11    }
12    void insert(int x) { // log
13        count[x]++;
14        if (count[x] == 1)
15            missing.erase(x);
16    }
17    void remove(int x) { // log
18        if (count[x] == 0) return;
19        count[x]--;
20        if (count[x] == 0)
21            missing.insert(x);
22    }
23    int get_mex() { // 1
24        return *missing.begin();
25    }
26 };
```

Remove Game

```
1 void solve() {
2     int n; cin >> n;
3     vector<int> v(n);
4     getv(v);
5     ll diff[n+1][n+1]{}; // mx diff p1-p2 on interval [l, r]
6     for (int l = n-1; ~l; l--) {
7         for (int r = 0; r < n; r++) {
8             // dp[l][r] = mx( v[l]-dp[l+1][r], v[r]-dp[l][r-1] )
9             if (l == r)
10                diff[l][r] = v[l];
11             else diff[l][r] = max<ll>(v[l] - diff[l+1][r], v[r]-diff[l][r-1]);
12         }
13     }
14
15     cout << (accumulate(all(v), 0ll) + diff[0][n-1])/2;
16 }
```

K-th Balanced Bracket Sequence

```
1 //O(n^2)
2 string kth_balanced(int n, int k) {
3     vector<vector<int>> d(2*n+1, vector<int>(n+1, 0));
```

```

4      d[0][0] = 1;
5      for (int i = 1; i <= 2*n; i++) {
6          d[i][0] = d[i-1][1];
7          for (int j = 1; j < n; j++)
8              d[i][j] = d[i-1][j-1] + d[i-1][j+1];
9          d[i][n] = d[i-1][n-1];
10     }
11     string ans;
12     int depth = 0;
13     for (int i = 0; i < 2*n; i++) {
14         if (depth + 1 <= n && d[2*n-i-1][depth+1] >= k)
15             {
16                 ans += '(';
17                 depth++;
18             } else {
19                 ans += ')';
20                 if (depth + 1 <= n)
21                     k -= d[2*n-i-1][depth+1];
22                 depth--;
23             }
24     }
25     return ans;
26 }

```

Next Balanced Sequence

```

1  bool next_balanced_sequence(string & s) { // O(n)
2      int n = s.size();
3      int depth = 0;
4      for (int i = n - 1; i >= 0; i--) {
5          if (s[i] == '(')
6              depth--;
7          else
8              depth++;
9          if (s[i] == '(' && depth > 0) {
10             depth--;
11             int open = (n - i - 1 - depth) / 2;
12             int close = n - i - 1 - open;
13             string next = s.substr(0, i) + ')' +
14                 string(open, '(') + string(close, ')');
15             s.swap(next);
16             return true;
17         }
18     }
19     return false;
20 }

```

9. Geometry

Geometry Notes

1 Generate 2 points on a line

2 // $ax + by + c = 0$

3 if($a == 0$){

4 $p1 = \{0, -1.0 * c/b\};$

5 $p2 = \{1, -1.0 * c/b\};$

6 }

7 else{

8 $p1 = \{-1.0 * c/a, 0\};$

9 $p2 = \{-1.0 * (c + b)/a, 1\};$

10 }

11

12

13 You're given 4 integers which are the

14 coefficients A B and C of the normal equation of the

15 straight line and a distance value R.

16

17 void solve() {

18 double a,b,c,r; cin >> a >> b >> c >> r;

19 double base = sqrt($a * a + b * b$);

20 $a /= base;$

21 $b /= base;$

22 $c /= base;$

23 cout << setprecision(15) << a << ' ' << b << ' ' << c + r << endl;

24 cout << setprecision(15) << a << ' ' << b << ' ' << c - r << endl;

25 }

26

27 Area of the sector without an angle = $(l * r) / 2$

28 The length of the arc $l = (\text{theta} / 360) * 2 * \pi * r$

29

30 The area of the parallelogram = the cross-product of

31 2 adjacent sides = 2 * area of the triangle made by 3 points.

32

33 Given 1 side of the Pythagorean Triangle ... Get the missing 2 sides :

34 ll n;

35 cin >> n;

36 if ($n == 1 || n == 2$)

37 cout << -1;

38 else if ($n \& 1$)

39 cout << $(n * n + 1) / 2$ << " " << $(n * n - 1) / 2$;

40 else

41 cout << $n * n / 4 + 1$ << " " << $n * n / 4 - 1$;

42

43 In triangle abc angle bac $\cos(\text{theta}) = (b^2 + c^2 - a^2) / (2 * b * c)$

44

45 n = number of sides of a regular polygon

46 S = side length of the polygon

47 ap = apothem the distance from the center of the polygon to the middle of any side

48 r = radius of the polygon which is the distance from the center of the polygon to any corner.

49 p = perimeter of the polygon

50

51 $p = S * n$

52 $ap = S / (2 * \tan(180/n)) = r * \cos(180/n)$

53 $r = S / (2 * \sin(180/n)) = ap / \cos(180/n)$

54 Area = $(p * ap) / 2$, $(S^2 * n) / (4 * \tan(180/n)) = ap^2 * n * \tan(180/n)$

```

55     = (r^2 * n * sin(360/n))/2
56
57 sin(2*theta) = 2 * sin(theta) * cos(theta)
58 cos(2*theta) = cos(theta)^2 - sin(theta)^2 = 2 * cos(theta)^2 - 1
59     = 1 - 2 * sin(2*theta)^2
60 sin(theta)^2 = (1 - cos(2 * theta))/2
61 cos(theta)^2 = (1 + cos(2 * theta))/2
62 tan(2*theta) = (2 * tan(theta)) / (1 - tan(theta)^2)
63
64 Circle intersection r1,r2,d where r1 >= r2
65 If d = r1 + r2 they touch from outside
66 If d = r1 - r2 they touch from inside
67 If r1 - r2 < d < r1 + r2 they intersect in two points
68
69 Plane equation ax + by + cz + d = 0
70 AB = (Bx-Ax,By-Ay,Bz-Az)
71 AC = (Cx-Ax,Cy-Ay,Cz-Az)
72 AB x AC = (a,b,c)
73 a = (By-Ay)*(Cz-Az)-(Cy-Ay)*(Bz-Az)
74 b = (Bz-Az)*(Cx-Ax)-(Cz-Az)*(Bx-Ax)
75 c = (Bx-Ax)*(Cy-Ay)-(Cx-Ax)*(By-Ay)
76 d = -(a*Ax+b*Ay+c*Az)
77
78 // Checks if four points lie in the same plane or not
79 bool samePlane(point a,point b,point c){
80     // a * (b x c) = volume = 0
81     return (a.dot(b.cross(c)) == 0);
82 }
83
84 void solve() {
85     vector<point>v(4);
86     for (int i = 0; i < 4; ++i) {
87         cin >> v[i].x >> v[i].y >> v[i].z;
88     }
89     for (int i = 0; i < 4; ++i) {
90         v[i].x -= v[3].x;
91         v[i].y -= v[3].y;
92         v[i].z -= v[3].z;
93     }
94     cout << (samePlane(v[0],v[1],v[2])) ? "YES":"NO" << endl;
95 }

```

Geometry (X, Y, dot, cross)

```

1  /*
2  conj(a) -> a.imag() *= -1
3  abs(point) distance between (0,0) to this point
4  norm(point) squared magnitude -> real^2 + imag^2
5  hypot(x, y) -> sqrt(x^2 + y^2)
6  arg(vector) angle between this vector and x-axis
7  clamp(a, l, r) == min(r, max(l, a))
8  polar(rho, theta) -> make vector with length rho and angle theta
9  internal angle = (n - 2) * 180 / n
10 number of diagonals n * (n - 3) / 2
11 Area(p) = internal_points_cnt + (boundary_points/2) - 1
12 boundary_point in vector = gcd(|x2-x1|, |y2-y1|) + 1

```

```

13  line have infinity point, segment have to end points
14  vector(x, y) perpendicular to vector(-y, x) and (y, -x)
15  */
16
17  using ll = int64_t;
18
19  using ld = double;
20  using pt = complex<ld>;
21
22  const ll INF = 7e18;
23  const ld EPS = 1e-9;
24  const ld PI = acos(-1);
25
26  #define X real()
27  #define Y imag()
28
29  #define dot(a, b) (conj(a) * (b)).X
30  #define cross(a, b) (conj(a) * (b)).Y
31
32  int sign(ld x) {
33      return (x > EPS) - (x < -EPS);
34  }
35
36  struct compX{
37      bool operator()(pt a, pt b) const {
38          return a.X != b.X ? a.X < b.X : a.Y < b.Y;
39      }
40  };
41  struct compY{
42      bool operator()(pt a, pt b) const {
43          return a.Y != b.Y ? a.Y < b.Y : a.X < b.X;
44      }
45  };
46
47  // ===== line, segment =====
48
49  // projection of pt p onto line ab
50  pt project(pt a, pt b, pt p) {
51      pt ab = b - a;
52      return a + ab * dot(p - a, ab) / norm(ab);
53  }
54
55  // works for any orientation
56  bool onSegment(pt a, pt b, pt p) {
57      return sign(cross(b - a, p - a)) == 0 &&
58             sign(dot(p - a, p - b)) <= 0;
59  }
60
61  // ccw: >0 left, <0 right, =0 collinear
62  int ccw(pt a, pt b, pt c) {
63      return sign(cross(b - a, c - a));
64  }
65
66  // works for any pts
67  ld distanceToLine(pt a, pt b, pt p) {
68      return fabs(cross(b - a, p - a)) / abs(b - a);
69  }

```



```

70 // works for any line
71 ld distanceToLine(ld A, ld B, ld C, pt p) {
72     return fabs(A*p.X + B*p.Y + C) / abs(pt(A, B));
73 }
74
75 // works for any pts
76 ld distanceToSegment(pt a, pt b, pt p) {
77     if (dot(b - a, p - a) < 0) return abs(p - a);
78     if (dot(a - b, p - b) < 0) return abs(p - b);
79     return distanceToLine(a, b, p);
80 }
81
82 // works for intersecting lines (not parallel)
83 pt lineIntersect(pt a, pt b, pt c, pt d) {
84     pt ab = b - a, cd = d - c;
85     return a + ab * (cross(c - a, cd) / cross(ab, cd));
86 }
87
88 // works for all segments (returns intersection pt if exists)
89 bool segmentsIntersect(pt a, pt b, pt c, pt d, pt &inter) {
90     int d1 = ccw(a, b, c), d2 = ccw(a, b, d);
91     int d3 = ccw(c, d, a), d4 = ccw(c, d, b);
92
93     if(d1 * d2 < 0 && d3 * d4 < 0)
94         return inter = lineIntersect(a, b, c, d), true;
95
96     if(d1 == 0 && onSegment(a, b, c)) return inter = c, true;
97     if(d2 == 0 && onSegment(a, b, d)) return inter = d, true;
98     if(d3 == 0 && onSegment(c, d, a)) return inter = a, true;
99     if(d4 == 0 && onSegment(c, d, b)) return inter = b, true;
100
101     return false;
102 }
103
104 // works for any triangle
105 ld triangleArea(pt a, pt b, pt c) {
106     return 0.5 * fabs(cross(b - a, c - a));
107 }
108
109 bool ptInTriangle(pt a, pt b, pt c, pt p) {
110     ld s1 = cross(b - a, p - a);
111     ld s2 = cross(c - b, p - b);
112     ld s3 = cross(a - c, p - c);
113     return (sign(s1) >= 0 && sign(s2) >= 0 && sign(s3) >= 0) ||
114           (sign(s1) <= 0 && sign(s2) <= 0 && sign(s3) <= 0);
115 }
116
117 // angle abc in radians
118 ld angle_abc(pt a, pt b, pt c) {
119     return acos(clamp<ld>(dot(a - b, c - b) / (abs(a - b) * abs(c - b)), -1, 1));
120 }
121
122 // ===== Circles =====
123
124 pair<ld, pt> findCircle(pt a, pt b, pt c) {

```

```

126     pt m1 = (a + b) / 2.0, m2 = (b + c) / 2.0;
127     pt ab = b - a, bc = c - b;
128     pt center = lineIntersect(m1, m1 + pt(-ab.Y, ab.X),
129                               m2, m2 + pt(-bc.Y, bc.X));
130     return {abs(center - a), center};
131 }
132
133 vector<pt> lineCircleIntersect(pt a, pt b, pt center, ld r) {
134     pt ab = b - a, ao = center - a;
135     pt proj = a + ab * dot(ao, ab) / norm(ab);
136     ld d = abs(proj - center);
137     if (d > r + EPS) return {};
138     if (abs(d - r) < EPS) return {proj};
139     ld h = (ld)sqrtl(r*r - d*d);
140     pt dir = ab / abs(ab);
141     return {proj + dir * h, proj - dir * h};
142 }
143
144 // in 0, 1, 2 pts
145 vector<pt> circleCircleIntersect(pt c1, ld r1, pt c2, ld r2) {
146     ld d = abs(c2 - c1);
147     if(d > r1 + r2 + EPS || d < abs(r1 - r2) - EPS) return {};
148     if(abs(d) < EPS && abs(r1 - r2) < EPS) return vector(3, c1); // infinity intersection
149
150     ld a = (r1*r1 - r2*r2 + d*d) / (2 * d), h2 = r1*r1 - a*a;
151     if (h2 < -EPS) return {};
152
153     pt dir = (c2 - c1) / d, p = c1 + dir * a;
154     if (abs(h2) < EPS) return {p};
155     ld h = sqrt(h2);
156     pt offset = dir * pt(0, 1) * h;
157     return {p + offset, p - offset};
158 }
159
160 pair<ld, pt> minimumEnclosingCircle(vector<pt> p) {
161     using circle = pair<ld, pt>;
162     shuffle(p.begin(), p.end(), mt19937(random_device{}()));
163     auto contains = [](circle c, const vector<pt>& pts) {
164         return all_of(pts.begin(), pts.end(),
165                       [&](auto p) {return abs(p - c.second) <= c.first + EPS;});
166     };
167     auto circleFrom2 = [](pt a, pt b) {
168         pt c = (a + b) / 2.0;
169         return circle{abs(a - c), c};
170     };
171     auto circleFrom3 = [](pt a, pt b, pt c) {
172         pt ab = (a + b) / 2.0, ac = (a + c) / 2.0;
173         pt ab_perp = (b - a) * pt(0, 1), ac_perp = (c - a) * pt(0, 1);
174         pt o = lineIntersect(ab, ab + ab_perp, ac, ac + ac_perp);
175         return circle{abs(o - a), o};
176     };
177     vector<pt> R;
178     function<circle(int)> welzl = [&](int n) -> circle {
179         if (n == 0 || R.size() == 3) {
180             if (R.empty()) return {};
181             if (R.size() == 1) return {0, R[0]};

```

```

182         if (R.size() == 2) return circleFrom2(R[0], R[1]);
183         return circleFrom3(R[0], R[1], R[2]);
184     }
185     pt q = p[n - 1];
186     circle D = welzl(n - 1);
187     if (contains(D, {q})) return D;
188     R.push_back(q);
189     auto res = welzl(n - 1);
190     R.pop_back();
191     return res;
192 };
193 return welzl((int)p.size());
194 }
195
196 // ===== polygon =====
197
198 // works for any polygon (returns +1 for ccw, -1 for cw)
199 ld polygonSign(vector<pt>& p) {
200     ld area = 0;
201     int n = (int)p.size();
202     p.push_back(p[0]);
203     for(int i = 0; i < n; ++i) area += cross(p[i], p[i + 1]);
204     p.pop_back();
205     return sign(0.5 * area);
206 }
207
208 // works for any polygon (removes dups, enforces ccw order)
209 void normPolygon(vector<pt>& p) {
210     vector<pt> res;
211     for(auto i : p) if(res.empty() || abs(i - res.back()) > EPS)
212         res.push_back(i);
213
214     if(res.size() > 1 && abs(res.front() - res.back()) < EPS)
215         res.pop_back();
216
217     if(polygonSign(res) < 0) reverse(res.begin(), res.end());
218
219     p = res;
220 }
221
222 // works for simple polygons with integer coordinates
223 ll internalPointsCount(vector<pt>& p) {
224     ll A2 = 0, B = 0;
225     int n = (int)p.size();
226     p.push_back(p[0]);
227     for (int i = 0; i < n; ++i) {
228         pt a = p[i], b = p[i + 1];
229         A2 += ll(a.X * b.Y - a.Y * b.X);
230         B += __gcd((ll)abs(b.X - a.X), (ll)abs(b.Y - a.Y));
231     }
232     p.pop_back();
233     return (abs(A2) - B + 2) / 2;
234 }
235
236 // works for any polygon (cw or ccw, convex or not)
237 ld polygonArea(const vector<pt>& p) {

```

```

238     int n = (int)p.size();
239     ld area = 0;
240     for (int i = 0; i+1 < n; ++i)
241         area += cross(p[i], p[i + 1]);
242     area += cross(p.back(), p.front());
243     return fabsl(area) / 2.0;
244 }
245
246 // works for any polygon (cw or ccw, convex or not)
247 bool ptInPolygon(const vector<pt> &p, pt o) {
248     int in = 0, n = (int)p.size();
249     for (int i = 0; i+1 < n; ++i) {
250         pt a = p[i], b = p[i + 1];
251         if (onSegment(a, b, o)) return true;
252         if (a.Y > o.Y != b.Y > o.Y) {
253             ld x = a.X + (b.X - a.X) *
254                 (o.Y - a.Y) / (b.Y - a.Y);
255             if(x > o.X) in ^= 1;
256         }
257     }
258     {
259         pt a = p.back(), b = p.front();
260         if (onSegment(a, b, o)) return true;
261         if ((a.Y > o.Y) != (b.Y > o.Y)) {
262             ld x = a.X + (b.X - a.X) *
263                 (o.Y - a.Y) / (b.Y - a.Y);
264             if(x > o.X) in ^= 1;
265         }
266     }
267     return in;
268 }
269
270
271 // work for simple convex polygon
272 bool ptInConvex(vector<pt> &poly, pt p) {
273     int n = int(poly.size());
274     if(n == 1) return sign(abs(poly[0] - p)) == 0;
275     if(n == 2) return onSegment(poly[0], poly[1], p);
276
277     pt f = poly[0];
278
279     if(sign(cross(poly[1] - f, p - f)) < 0 ||
280        sign(cross(poly[n - 1] - f, p - f)) > 0) return false;
281
282     int l = 1, r = n - 1;
283     while(r > l + 1) {
284         int mid = (l + r) >> 1;
285         if(sign(cross(poly[mid] - f, p - f)) > 0) l = mid;
286         else r = mid;
287     }
288     return ptInTriangle(f, poly[l], poly[r], p);
289 }
290
291 // works for any simple polygon (cw or ccw)
292 pt polygonCentroid(const vector<pt>& p) {
293     ld A = 0, c;

```

```

294     pt C(0, 0);
295     int n = (int)p.size();
296     pt cur, nxt;
297     for (int i = 0; i+1 < n; ++i) {
298         cur = p[i], nxt = p[i + 1];
299         c = cross(cur, nxt);
300         A += c;
301         C += (cur + nxt) * c;
302     }
303     cur = p.back(), nxt = p.front();
304     c = cross(cur, nxt);
305     A += c;
306     C += (cur+nxt) * c;
307
308     A *= 0.5;
309     if (abs(A) < EPS) return C;
310     return C / (6.0 * A);
311 }

```